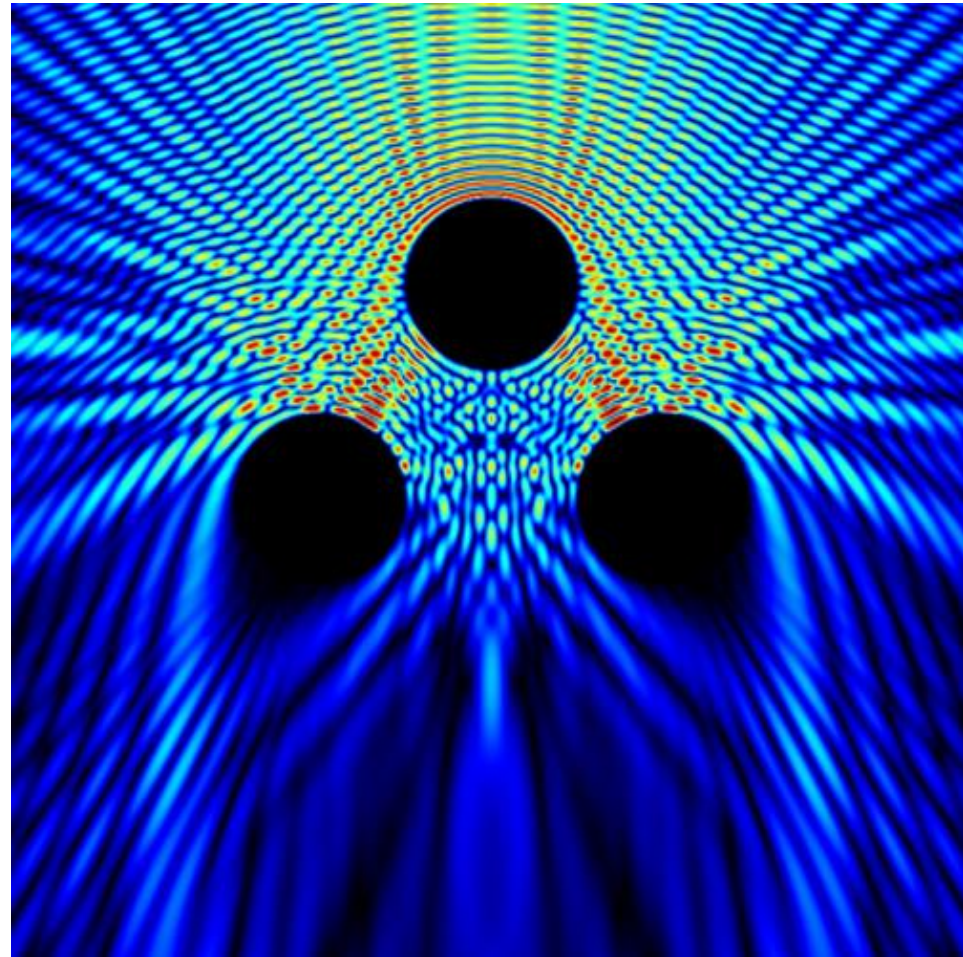


Engineering Electromagnetics

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Fundamentals of Electrostatics:

1. **Vector analysis,**
2. **Experimental law of Coulomb,**
3. **Electric field intensity,**
4. **Electric flux density,**
5. **Gauss' law,**
6. **Maxwell's First equation(Electrostatics),**
7. **divergence theorem,**
8. Energy expended in moving a point charge in an electric field,
9. The line integral,
10. Definition of potential difference and Potential,
11. The potential field of a point charge,
12. Potential gradient .

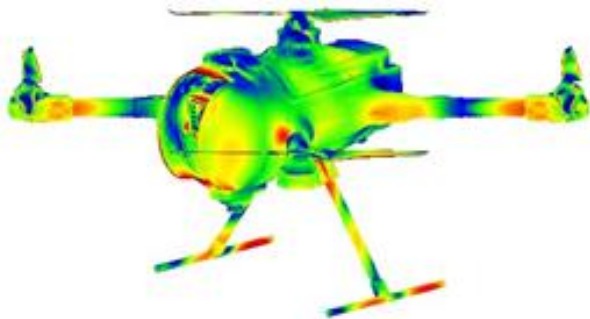
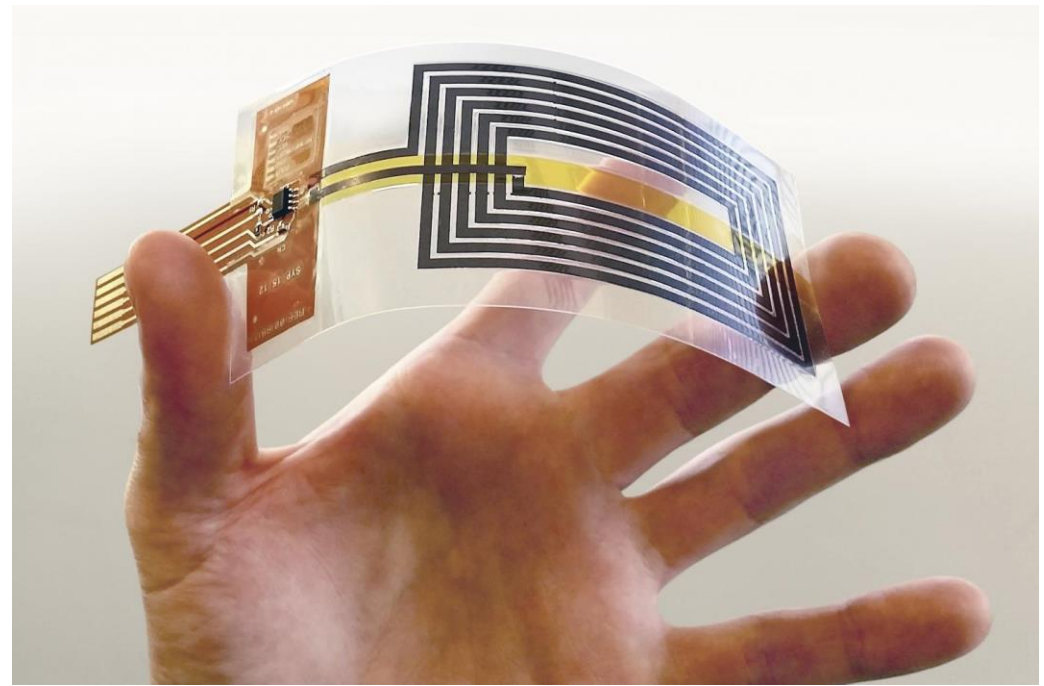
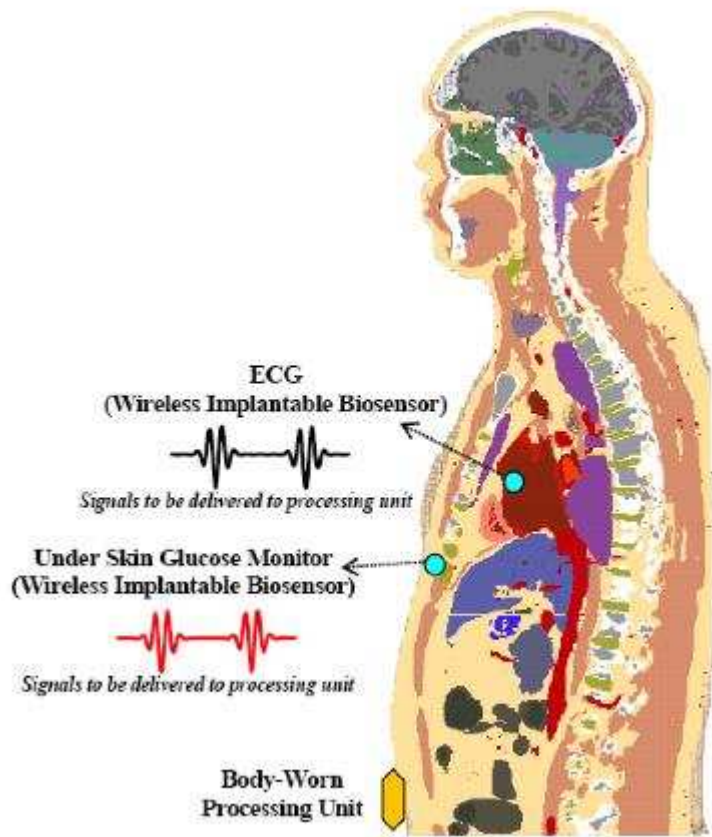
Electromagnetics:



- Branch of Physics/Electrical Engineering in which Electric and Magnetic phenomena are studied.
- Study of charges at rest and in motion.
- Electromagnetic principles are fundamental to the study of electrical engineering and physics.
- Electromagnetic theory is also indispensable to the understanding, analysis and design of various electrical, electromechanical and electronic systems.

Allied Disciplines:

Microwaves, Antennas, Electric Machines, Satellite Communications, Nuclear Research, Radar, Remote Sensing, EMI/C, Electromechanical energy conversion.



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minerals



contaminants



water



geothermal



geotechnical



slope stability

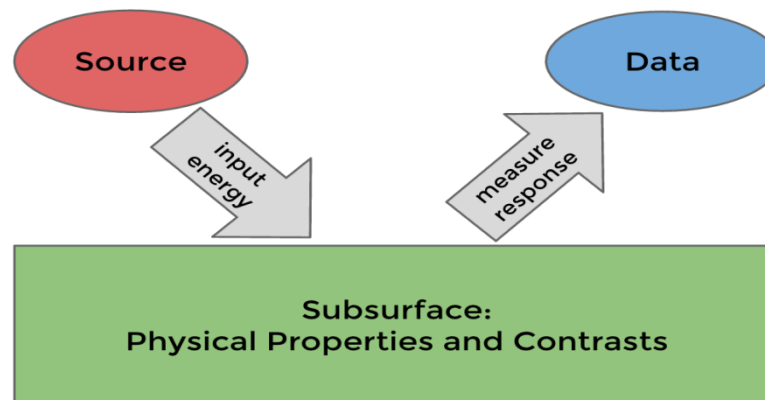


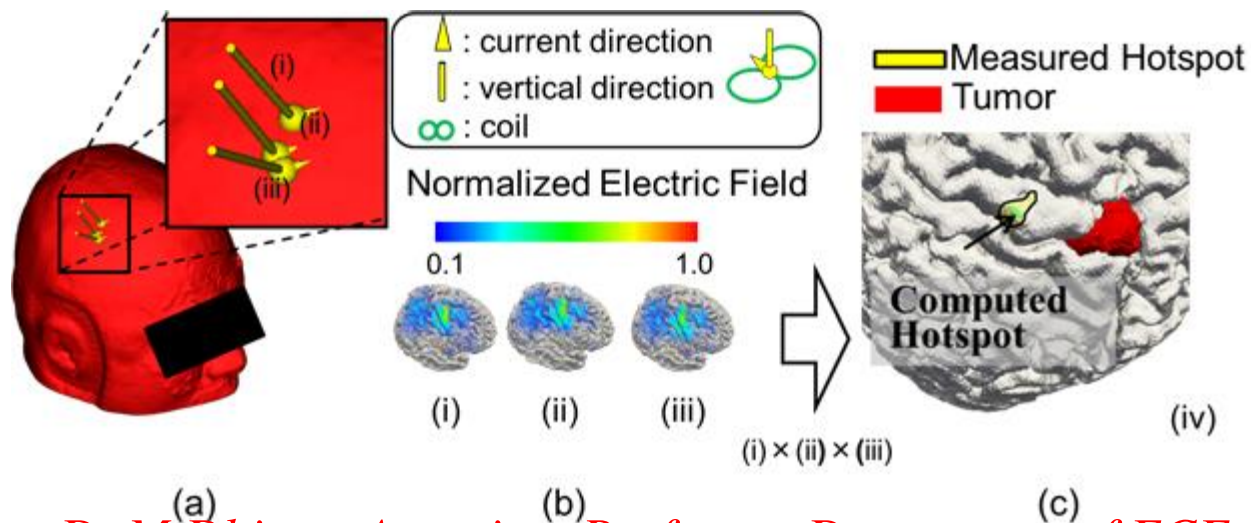
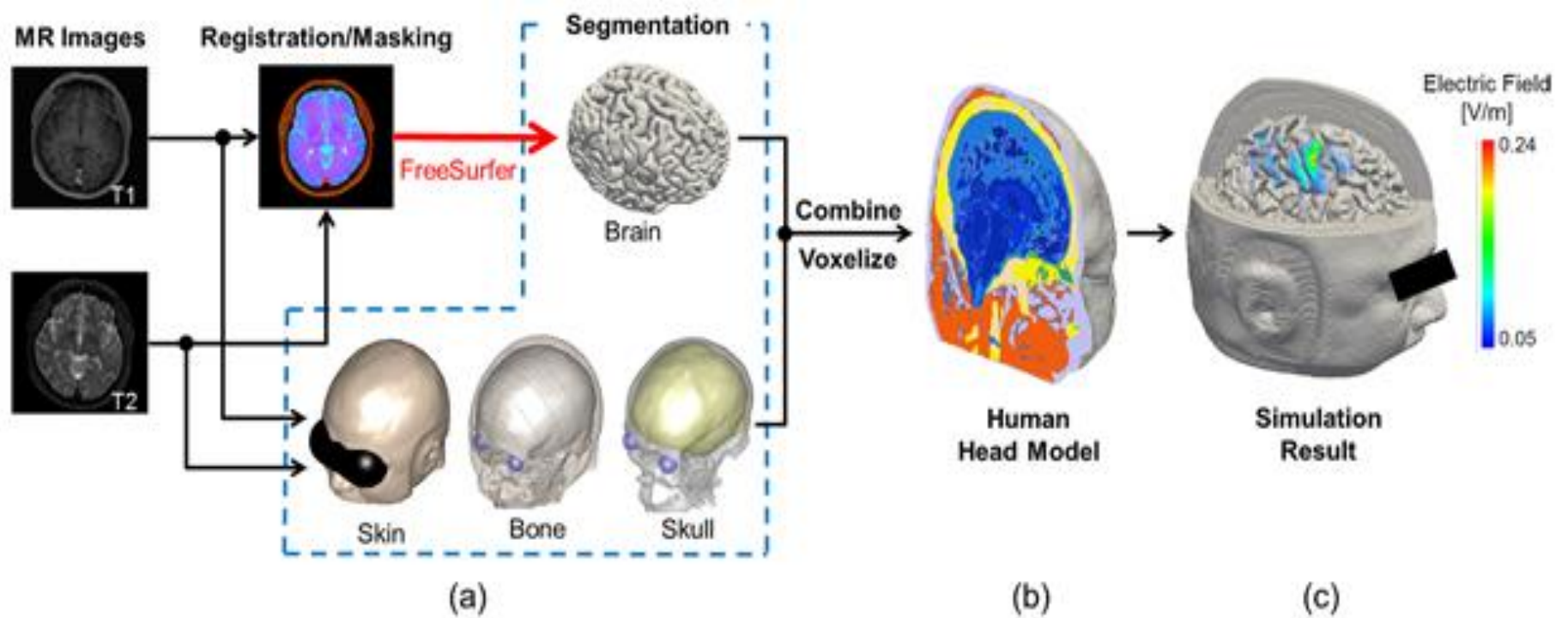
hydrocarbons



unexploded ordnance

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Measurement surface



Z

Y

X

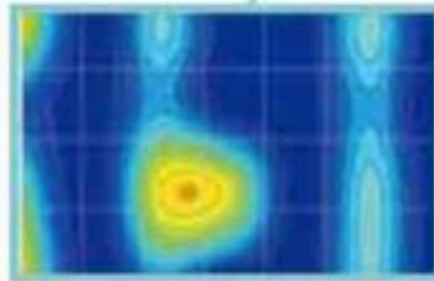
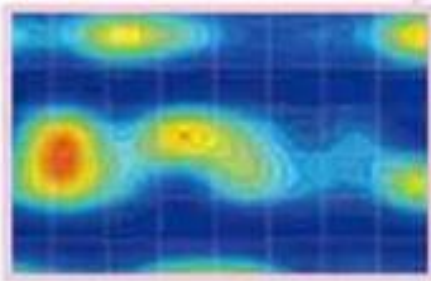
Measurement step

Virtual grid

Front

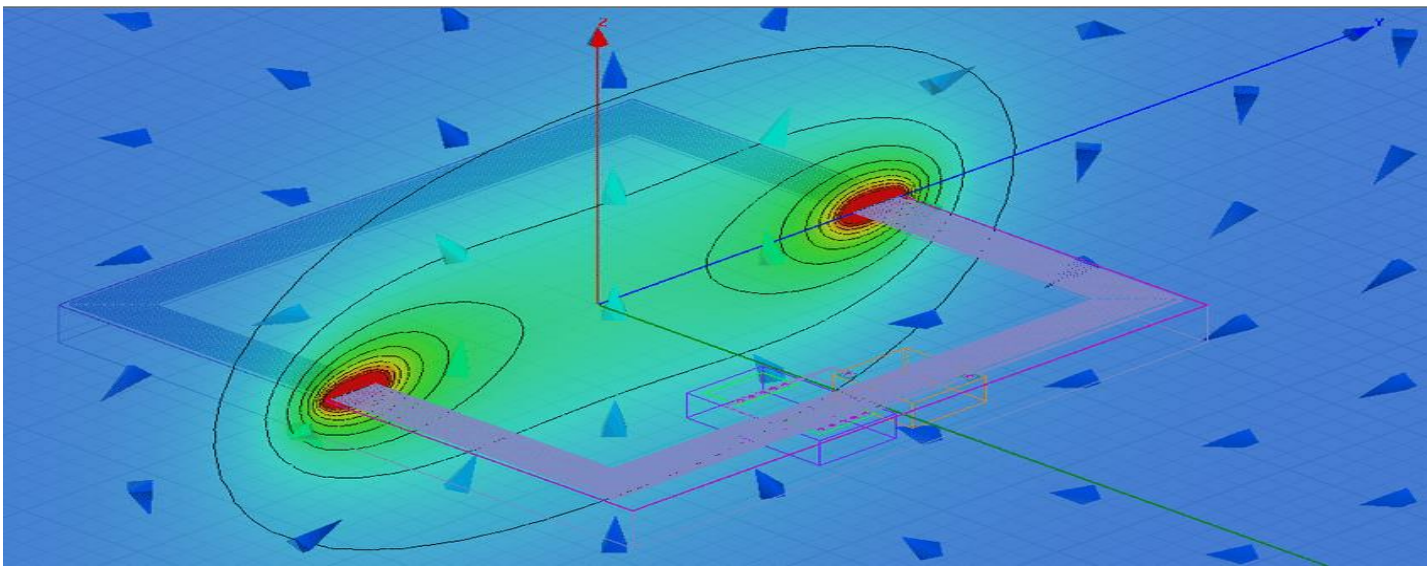
Back

The maximum measuring distance of the camera is 5 m. Can set an any grid and display noise distribution.



Extract tomographic images of any X-Y, X-Z, Y-Z.
Show the source of noise generation and noise distribution by visual

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Field – Region in which each point corresponds to some physical function.

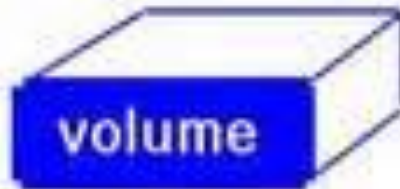
To quantify/measure the field three dimensional representation is required.

Three dimensional representation is made easy by vector analysis.

Vector analysis:

Scalar Quantities

length, area, volume
speed
mass, density
pressure
temperature
energy, entropy
work, power



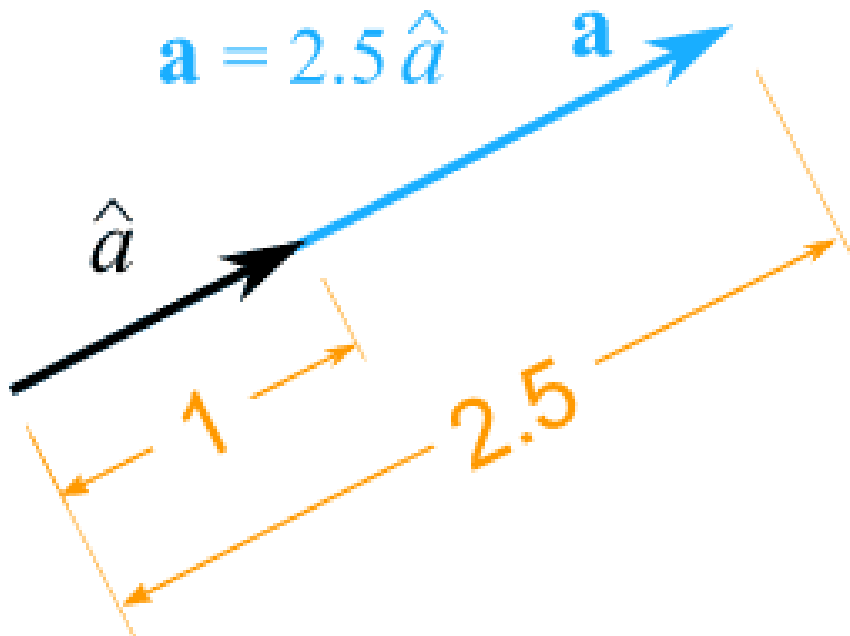
Vector Quantities

displacement, direction
velocity
acceleration
momentum
force
lift, drag, thrust
weight

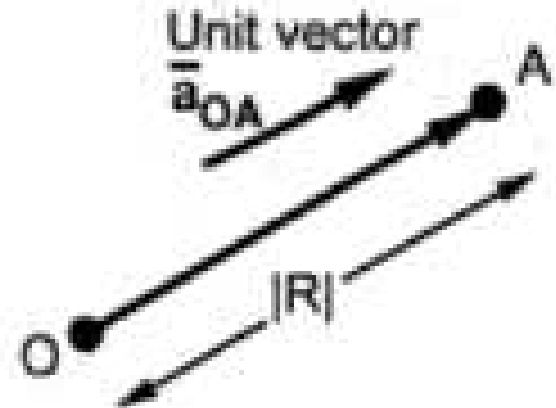


Unit Vector:

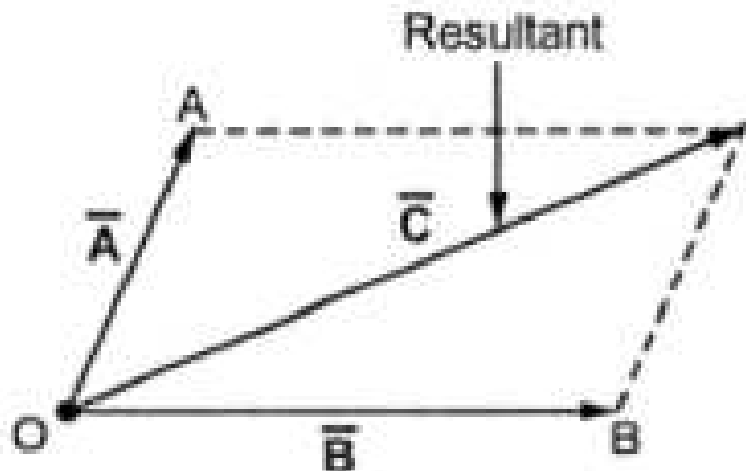
A unit vector has a function to indicate the direction.



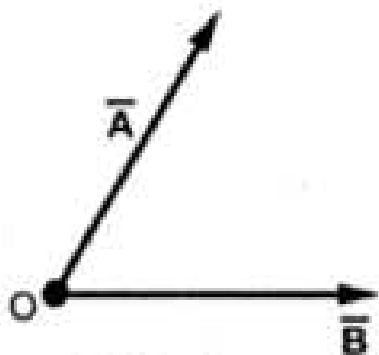
$$\overline{\mathbf{OA}} = |\overline{\mathbf{OA}}| \bar{\mathbf{a}}_{\mathbf{OA}} = R \bar{\mathbf{a}}_{\mathbf{OA}}$$



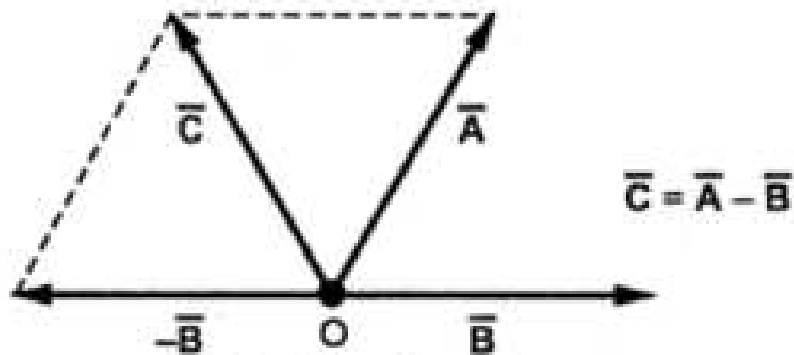
Vector Algebra:



Parallelogram rule for addition



(a) Vectors



Subtraction of vectors

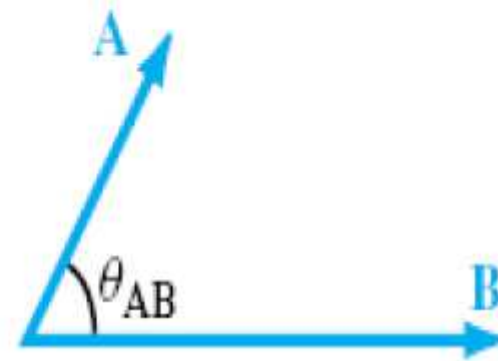
Dot Product & Cross Product:

Multiplication of Vectors:

When two vectors are multiplied, the result is either a scalar or a vector depending on how they are multiplied.

- Scalar (or dot) product $\vec{\mathbf{A}} \bullet \vec{\mathbf{B}}$
- Vector (or cross) product $\vec{\mathbf{A}} \times \vec{\mathbf{B}}$

- The dot product of two vectors, $\vec{\mathbf{A}}$ and $\vec{\mathbf{B}}$ is defined as the product of the magnitude of $\vec{\mathbf{A}}$, the magnitude of $\vec{\mathbf{B}}$ and the cosine of the **smaller angle** between them.



$$\vec{\mathbf{A}} \bullet \vec{\mathbf{B}} = |\mathbf{A}| |\mathbf{B}| \cos \theta_{AB}$$

$$\vec{\mathbf{A}} = A_x \mathbf{a}_x + A_y \mathbf{a}_y + A_z \mathbf{a}_z$$

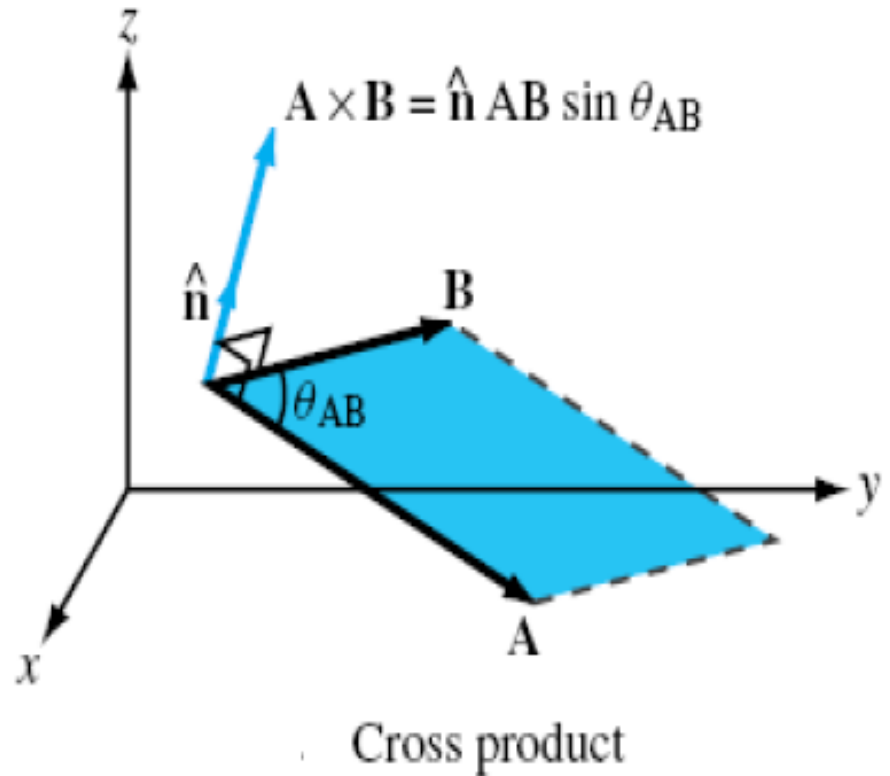
$$\vec{\mathbf{B}} = B_x \mathbf{a}_x + B_y \mathbf{a}_y + B_z \mathbf{a}_z$$

$$\vec{\mathbf{A}} \bullet \vec{\mathbf{B}} = A_x B_x + A_y B_y + A_z B_z$$

$$\mathbf{a}_x \bullet \mathbf{a}_y = \mathbf{a}_y \bullet \mathbf{a}_x = \mathbf{a}_x \bullet \mathbf{a}_z = \mathbf{a}_z \bullet \mathbf{a}_x = \mathbf{a}_y \bullet \mathbf{a}_z = \mathbf{a}_z \bullet \mathbf{a}_y = 0$$

$$\vec{A} \times \vec{B} = \hat{n} |\mathbf{A}| |\mathbf{B}| \sin \theta_{AB}$$

Direction of $\hat{n}$ is
perpendicular (90°) to
the plane containing A
and B



- Dot product obeys the following:

- Commutative Law $\vec{\mathbf{A}} \bullet \vec{\mathbf{B}} = \vec{\mathbf{B}} \bullet \vec{\mathbf{A}}$

- Distributive Law $\vec{\mathbf{A}} \bullet \vec{\mathbf{A}} = A^2 = |\mathbf{A}|^2$

$$\vec{\mathbf{A}} \bullet \left(\vec{\mathbf{B}} + \vec{\mathbf{C}} \right) = \vec{\mathbf{A}} \bullet \vec{\mathbf{B}} + \vec{\mathbf{A}} \bullet \vec{\mathbf{C}}$$

- Cross product obeys the following:

- It is not commutative $\vec{\mathbf{A}} \times \vec{\mathbf{B}} \neq \vec{\mathbf{B}} \times \vec{\mathbf{A}}$

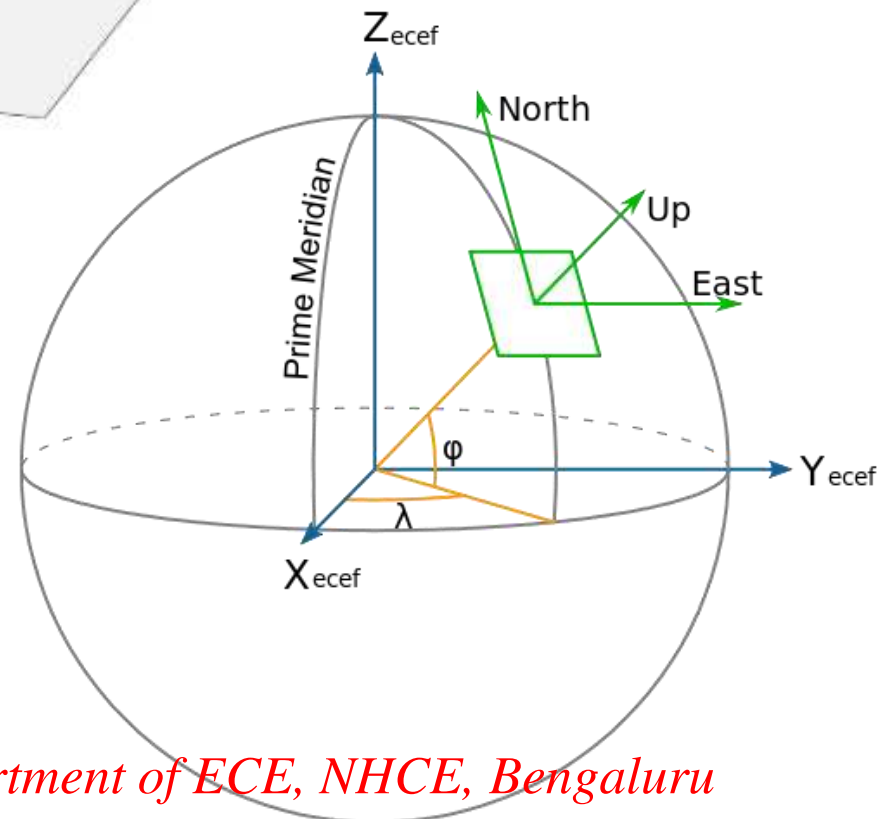
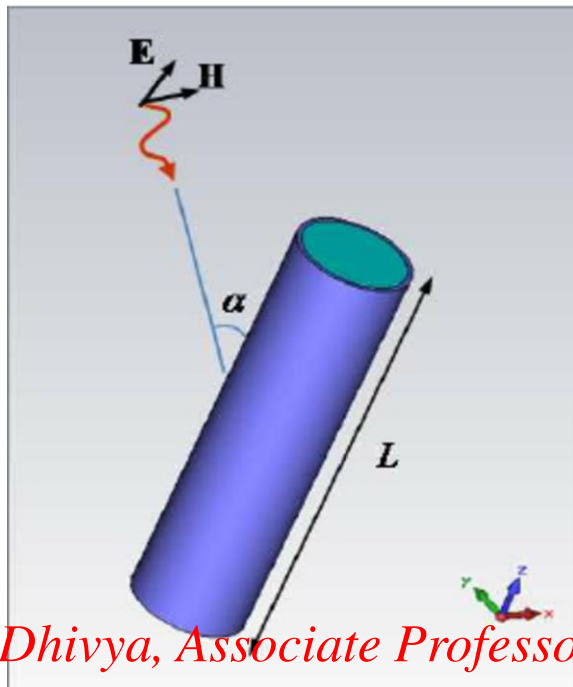
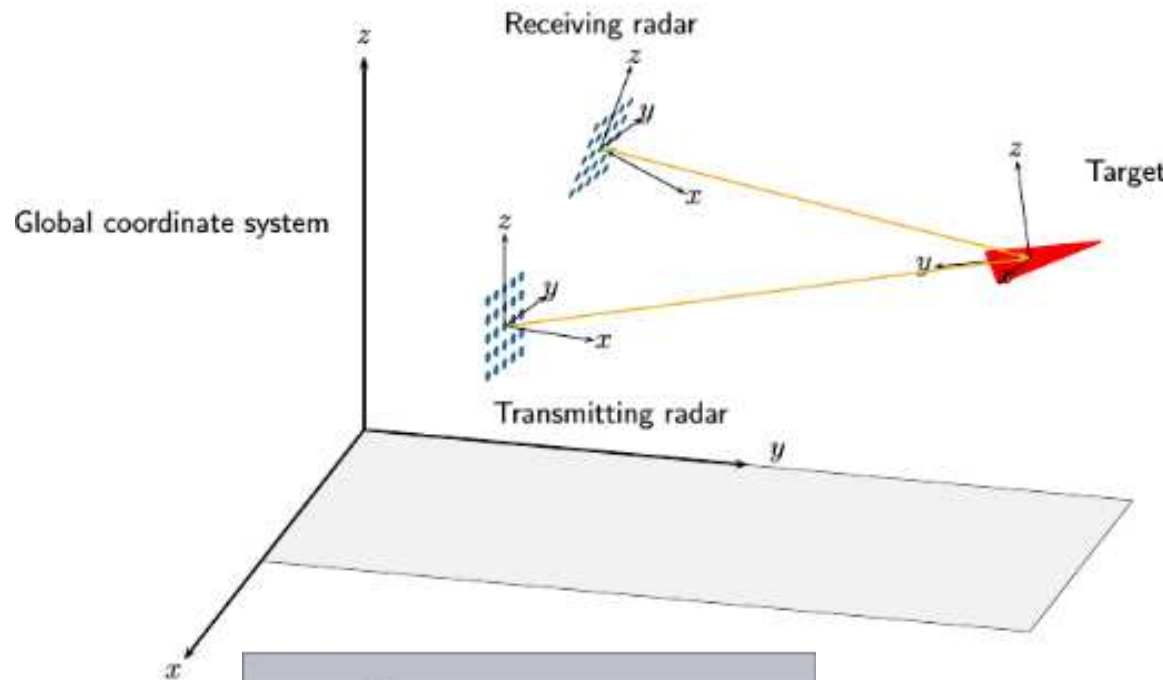
- It is not associative $\vec{\mathbf{A}} \times \left(\vec{\mathbf{B}} \times \vec{\mathbf{C}} \right) \neq \left(\vec{\mathbf{A}} \times \vec{\mathbf{B}} \right) \times \vec{\mathbf{C}}$

- It is distributive $\vec{\mathbf{A}} \times \left(\vec{\mathbf{B}} + \vec{\mathbf{C}} \right) \equiv \vec{\mathbf{A}} \times \vec{\mathbf{B}} + \vec{\mathbf{A}} \times \vec{\mathbf{C}}$

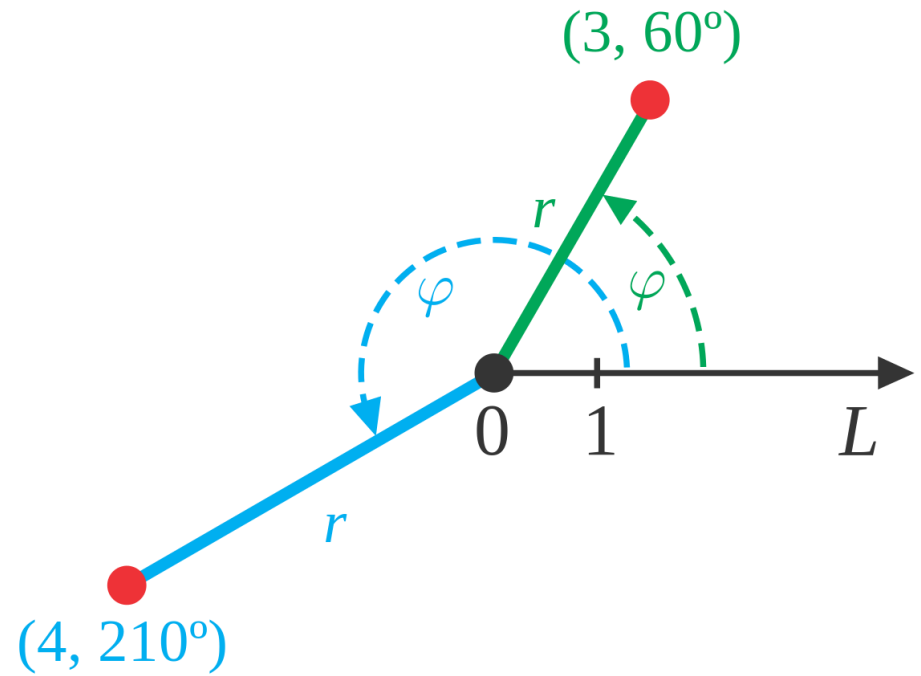
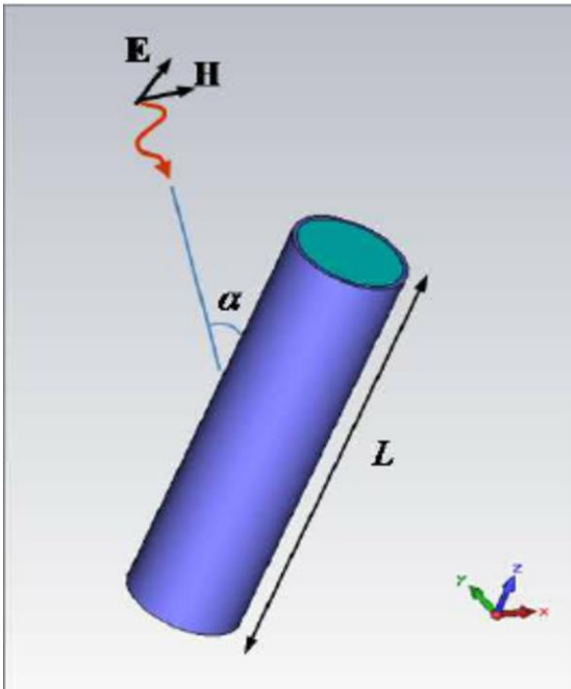
Coordinate system:

Aids accuracy in inferring the components and reference directions of any object/quantity

- Cartesian coordinates (x, y, z)
- Circular Cylindrical coordinates (ρ, ϕ, z)
- Spherical coordinates (r, θ, ϕ)

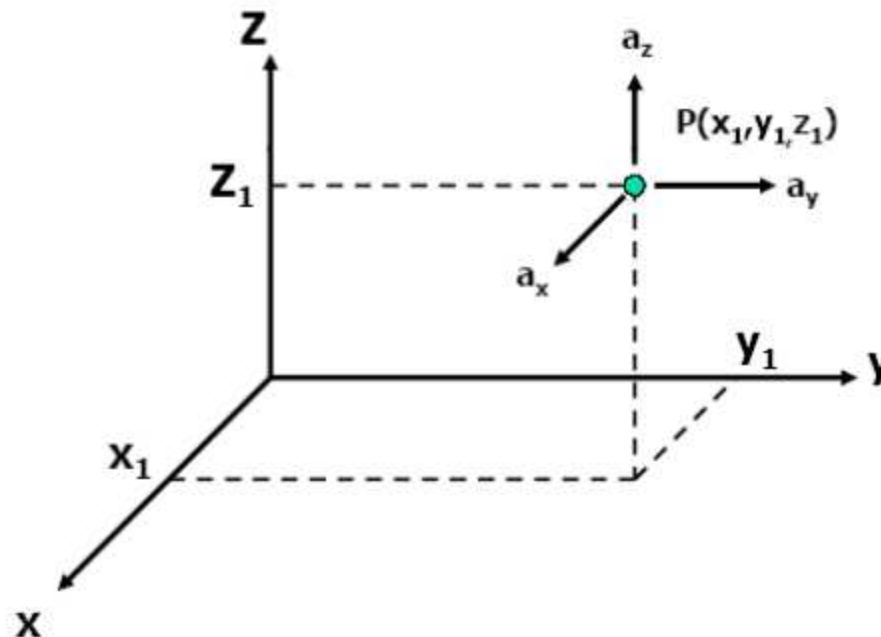


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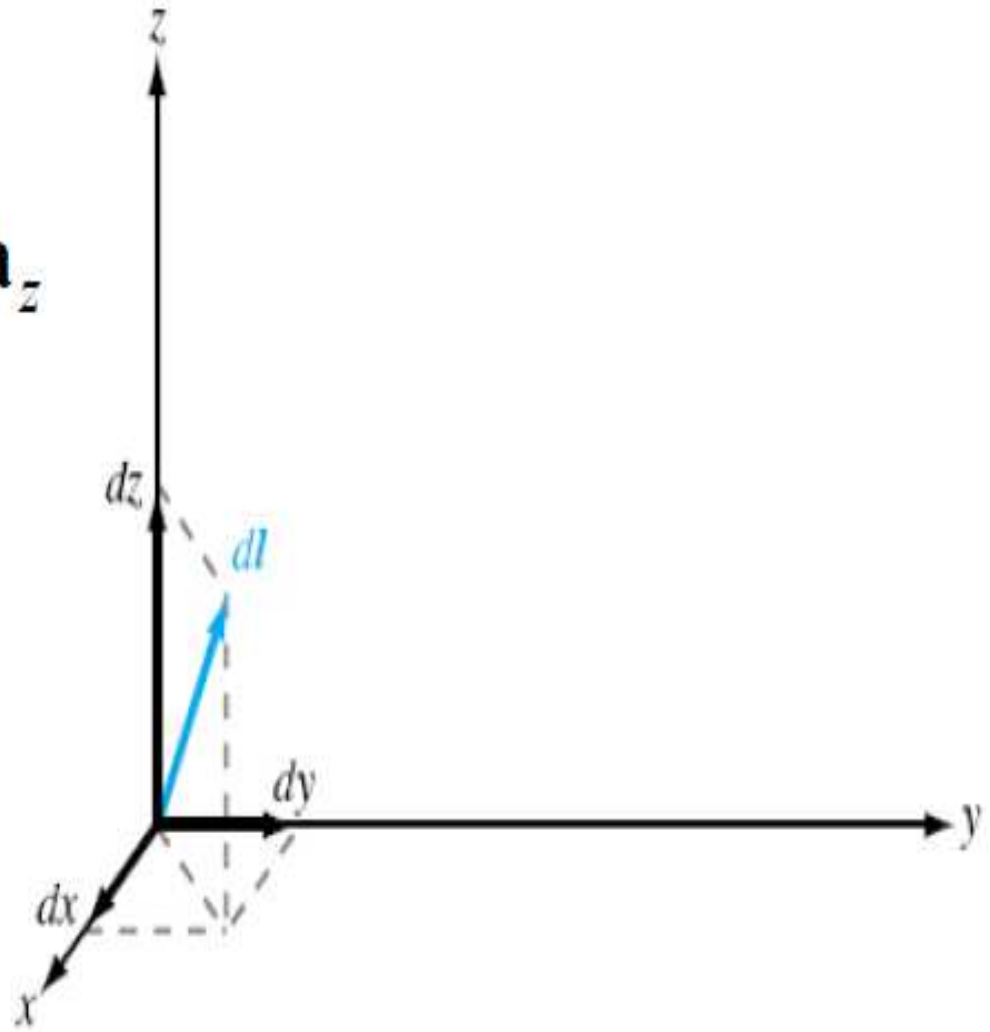
Cartesian Coordinates

- Unit vector of $\mathbf{a}_x, \mathbf{a}_y, \mathbf{a}_z$ in the direction of increasing coordinate value.

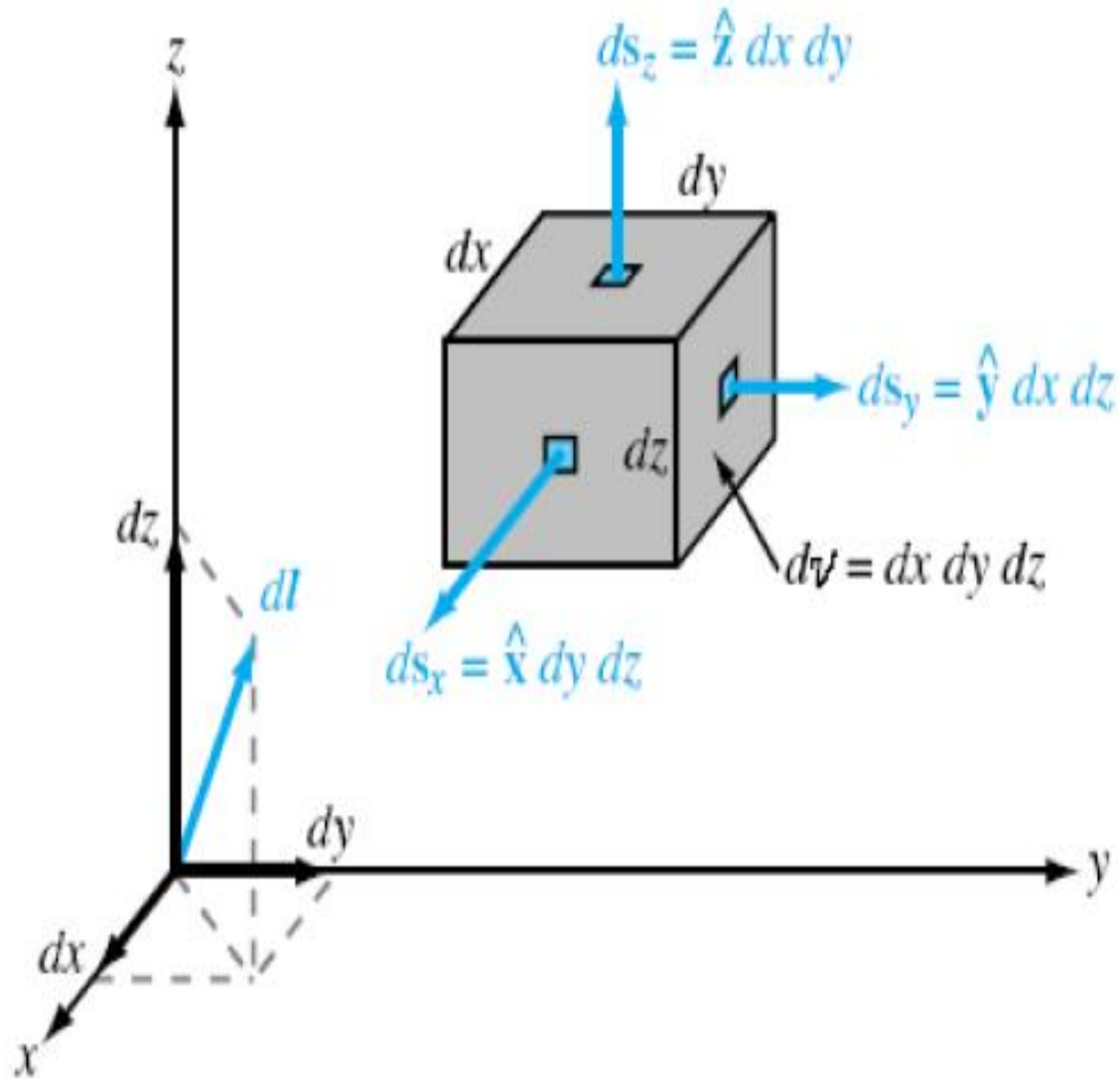


- Differential in Length

$$dL = dx\mathbf{a}_x + dy\mathbf{a}_y + dz\mathbf{a}_z$$



- Differential Surface

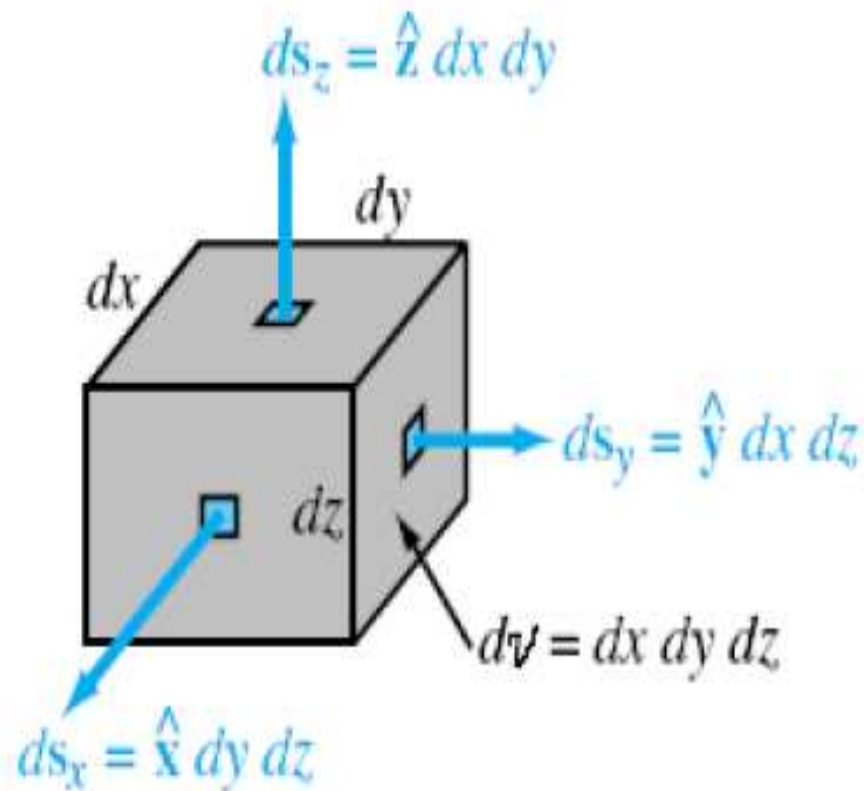


- Differential Surface

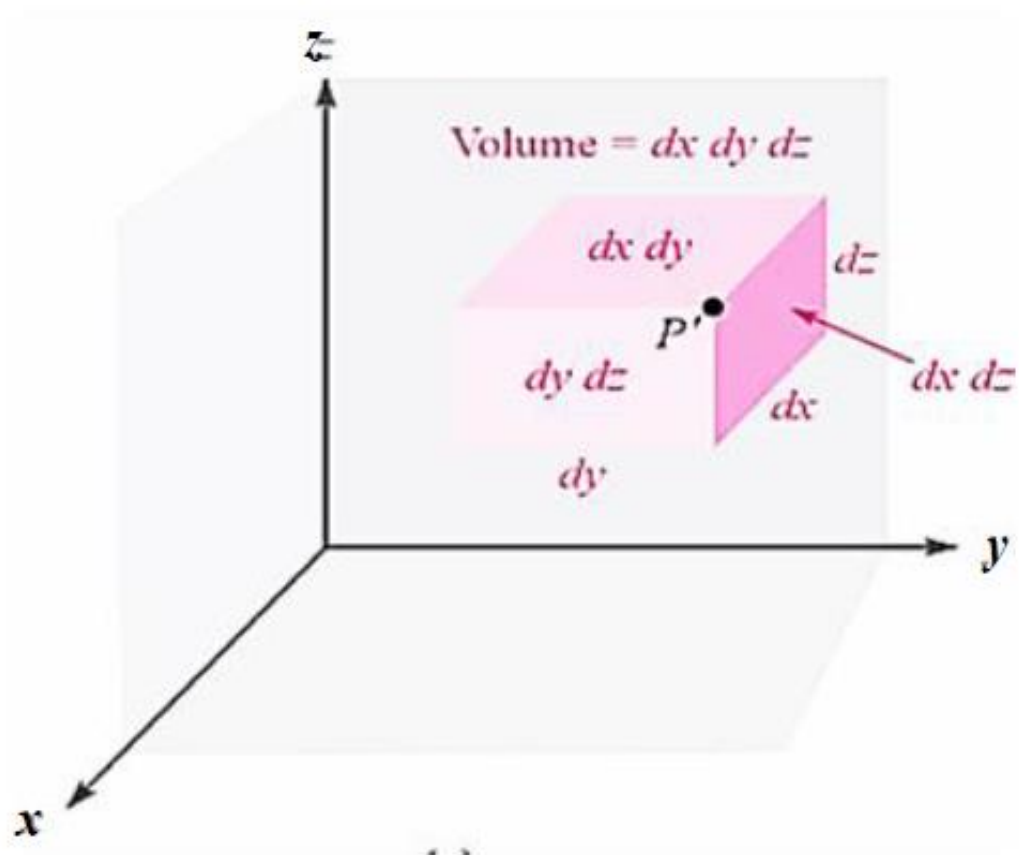
$$dS_x = dydz\mathbf{a}_x$$

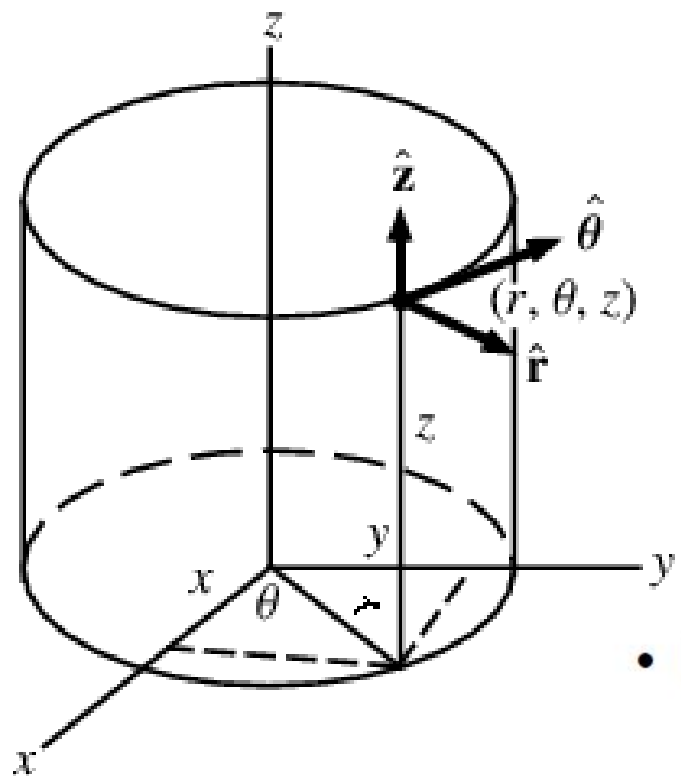
$$dS_y = dx dz\mathbf{a}_y$$

$$dS_z = dx dy\mathbf{a}_z$$

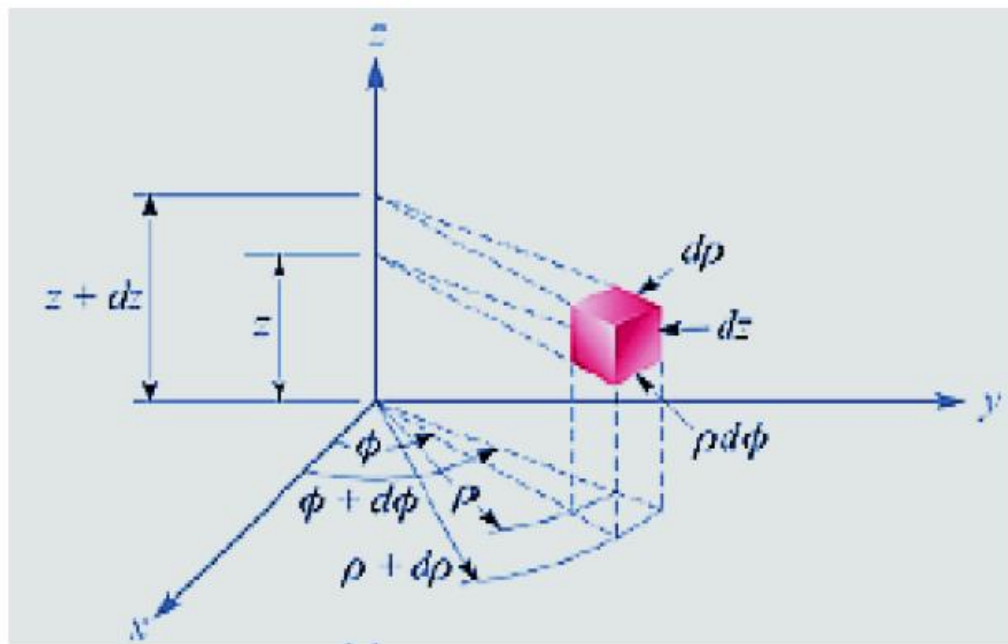


- Differential Volume $dV = dx dy dz$





- Differential in Length $dL = d\rho \mathbf{a}_\rho + \rho d\phi \mathbf{a}_\phi + dz \mathbf{a}_z$

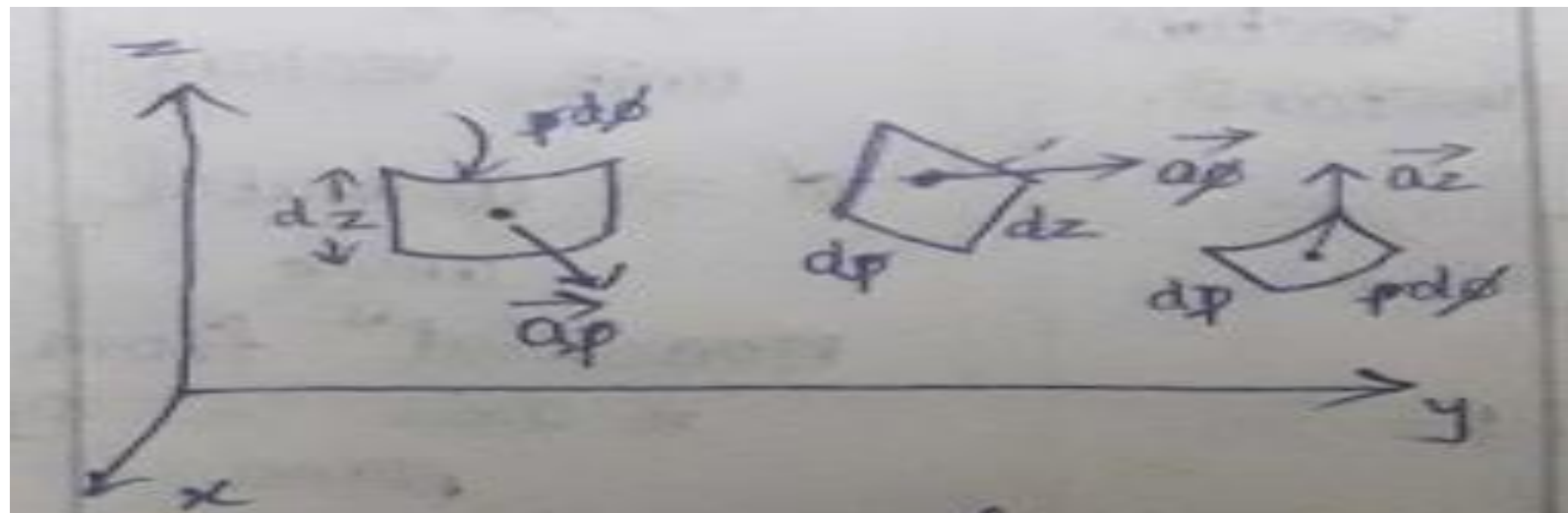
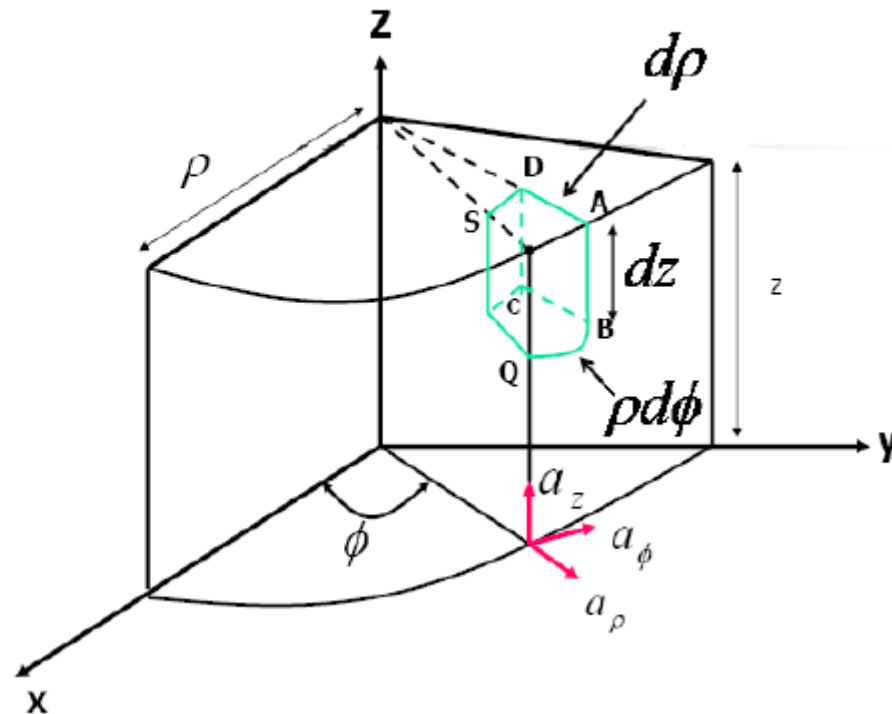


- Differential Surface

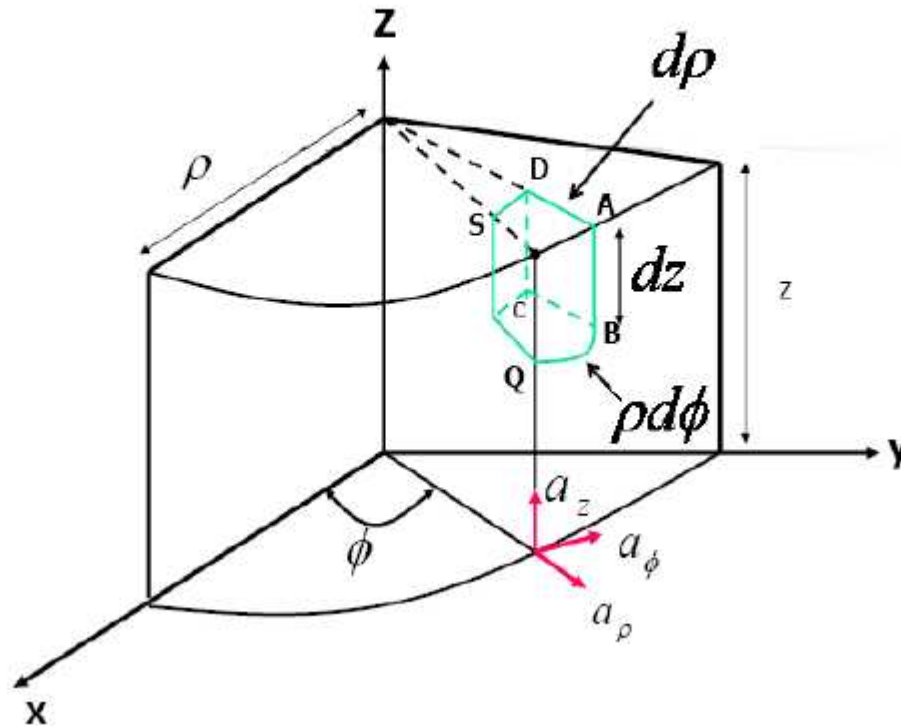
$$dS_\rho = \rho d\phi dz \mathbf{a}_\rho$$

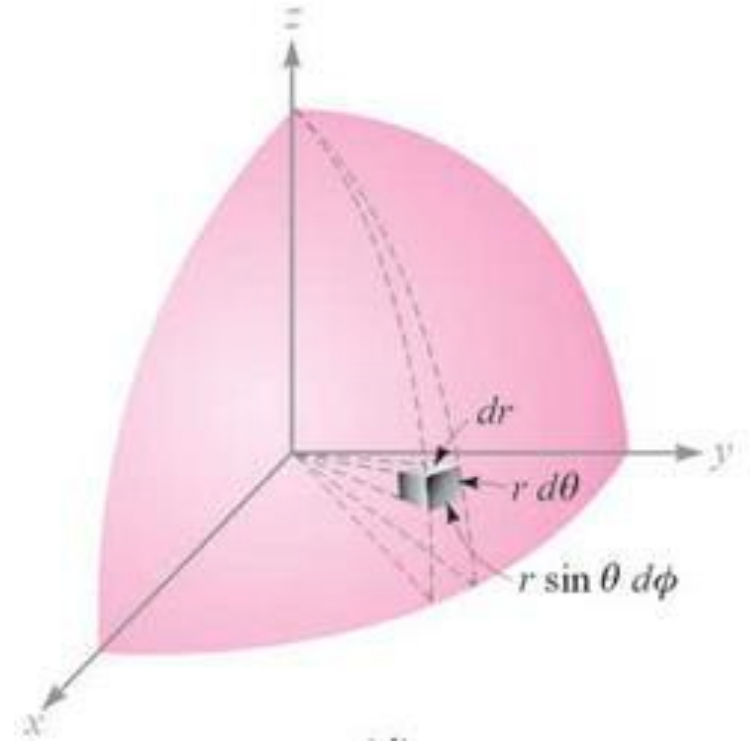
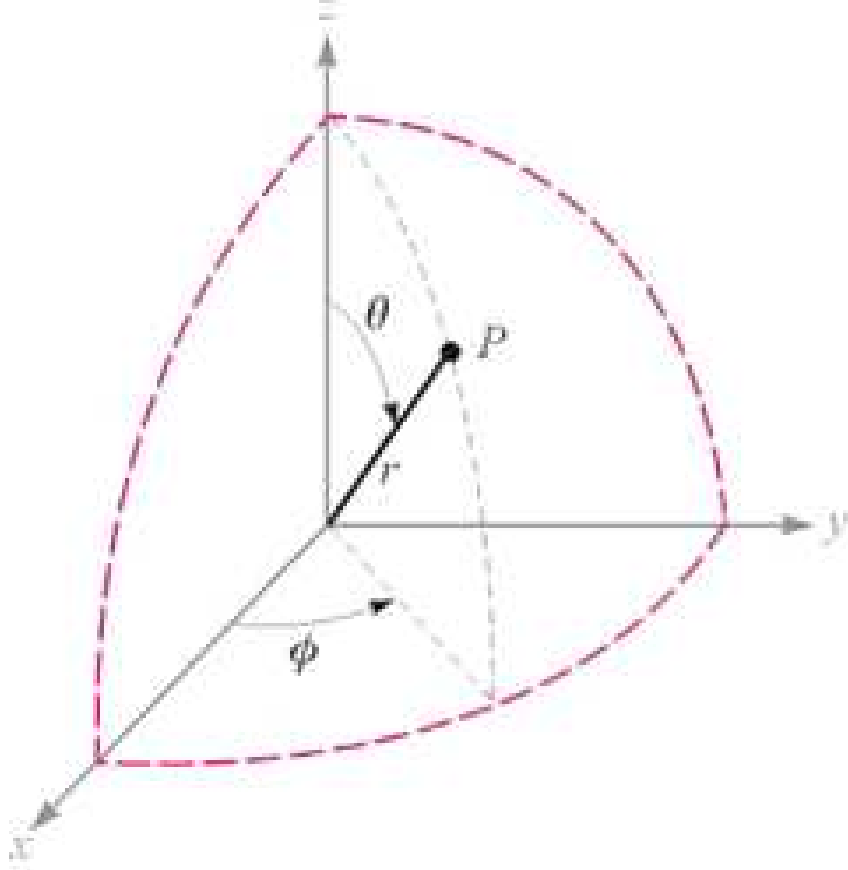
$$dS_\phi = d\rho dz \mathbf{a}_\phi$$

$$dS_z = \rho d\phi d\rho \mathbf{a}_z$$



- Differential volume $dV = \rho d\rho d\phi dz$





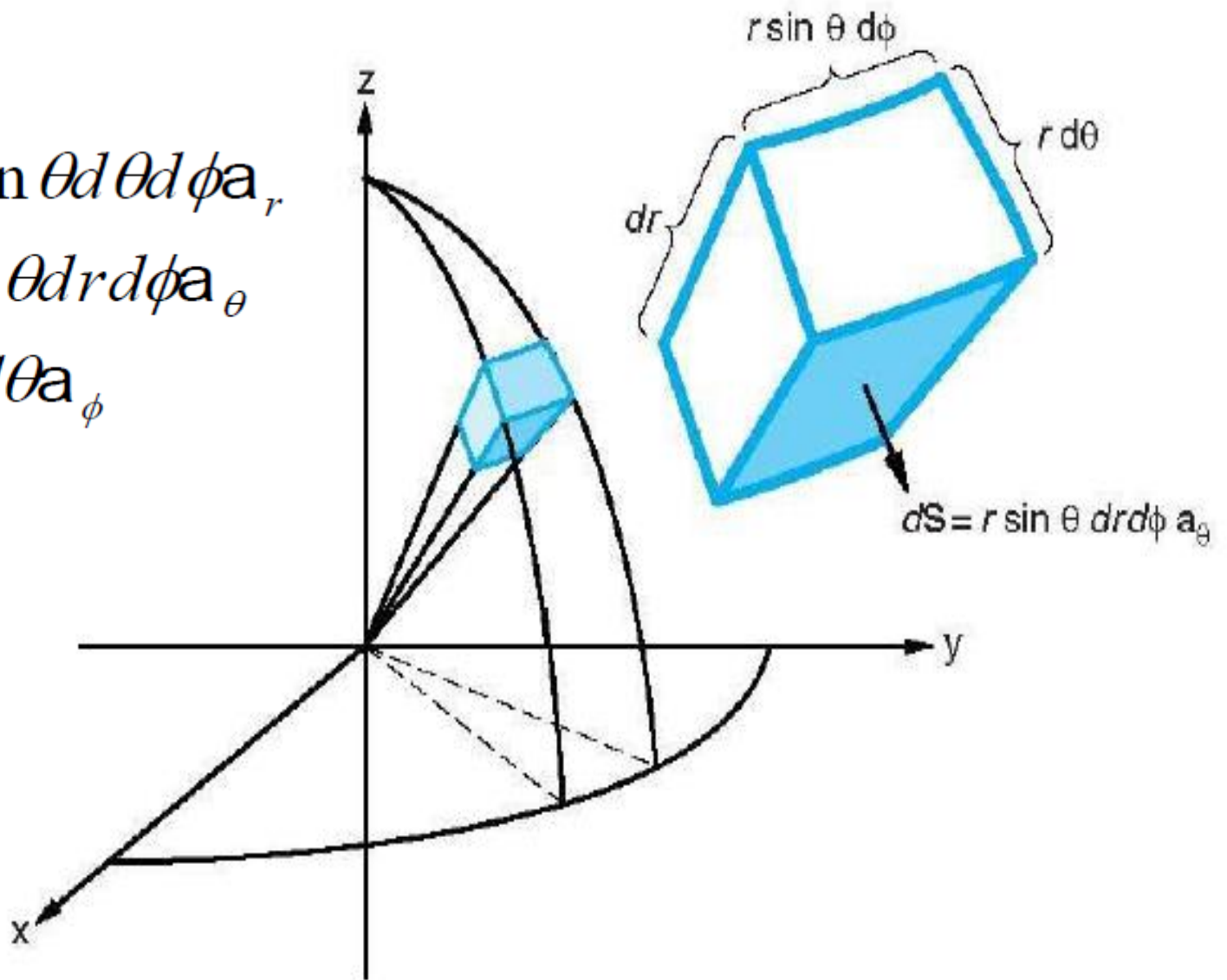
- Differential in length $dL = dr\mathbf{a}_r + r d\theta\mathbf{a}_\theta + r \sin \theta d\phi\mathbf{a}_\phi$

- Differential Surface

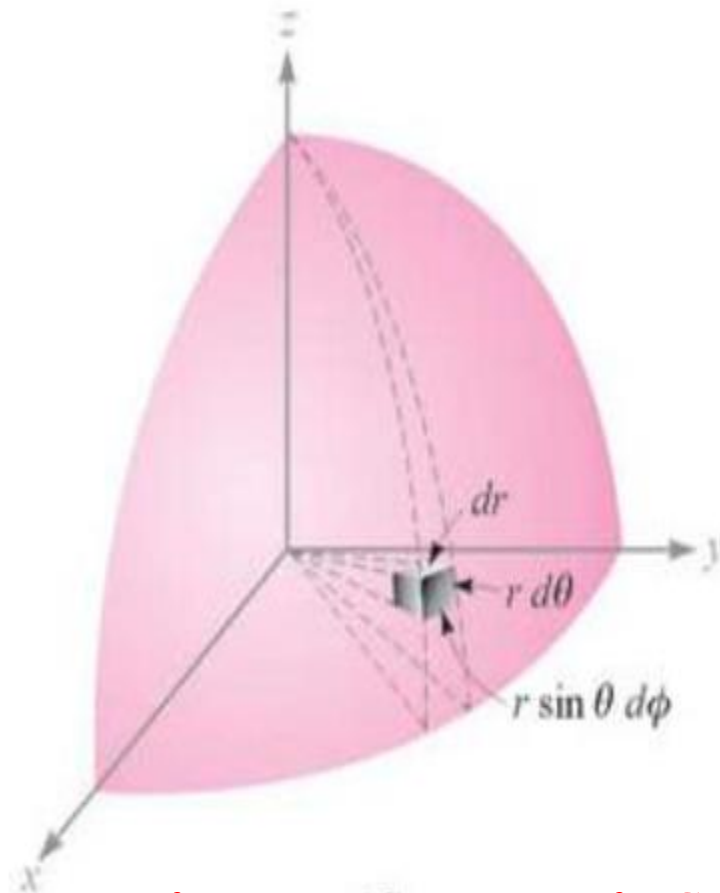
$$dS_r = r^2 \sin \theta d\theta d\phi \mathbf{a}_r$$

$$dS_\theta = r \sin \theta dr d\phi \mathbf{a}_\theta$$

$$dS_\phi = r dr d\theta \mathbf{a}_\phi$$

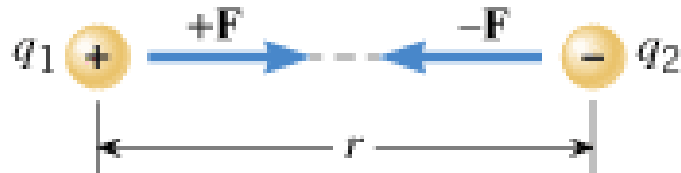


- Differential Volume $dV = r^2 \sin \theta dr d\theta d\phi$

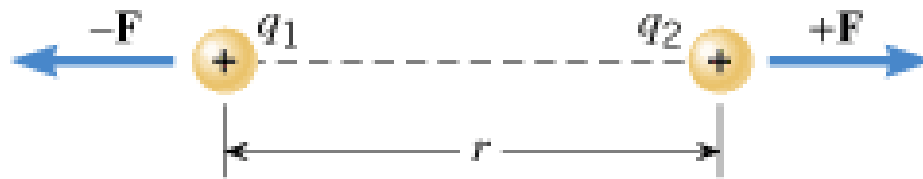


Electrostatics is a branch of physics that studies electric charges at rest.

Electrostatic forces are attractive or repulsive **forces** between particles that are caused by their electric charges. This **force** is also called the Coulomb **force** or Coulomb interaction.



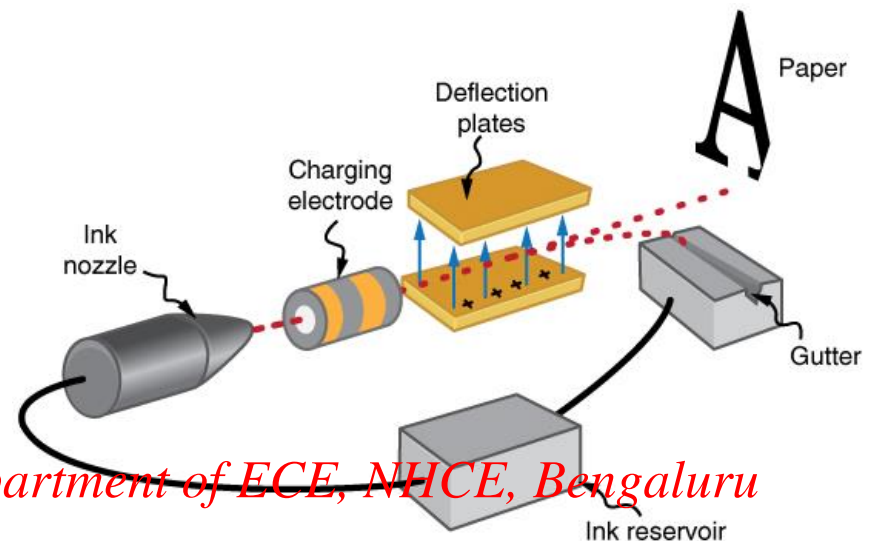
(a)



(b)

Applications:

LCD, Keyboard, Touch Pad, X-ray Machine, Spray Painting, Agriculture to sort seeds, Direct Spray to Plants, Air Cleaning etc.



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Experimental Law of Coulomb:

The Coulomb's law states that force between the two point charges Q_1 and Q_2 ,

1. acts along the line joining the two point charges.
2. is directly proportional to the product ($Q_1 Q_2$) of the two charges.
3. is inversely proportional to the square of the distance between them.

Application/Need:

The vector form of **Coulomb's law** is **important** as it helps us specify the direction of electric fields due to charges.

$$F \propto \frac{Q_1 Q_2}{R^2}$$

where

$Q_1 Q_2$ = Product of the two charges

R = Distance between the two charges

$$F = k \frac{Q_1 Q_2}{R^2}$$

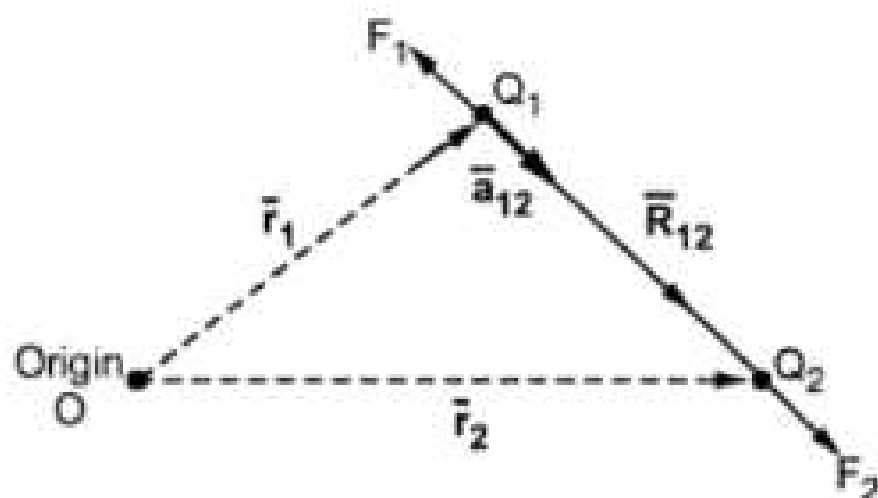
$$k = \frac{1}{4\pi \epsilon}$$

k = Constant of proportionality

For the **free space** or **vacuum**, the relative permittivity $\epsilon_r = 1$, hence

$$\epsilon = \epsilon_0$$

$$\therefore F = \frac{1}{4\pi\epsilon_0} \frac{Q_1 Q_2}{R^2}$$



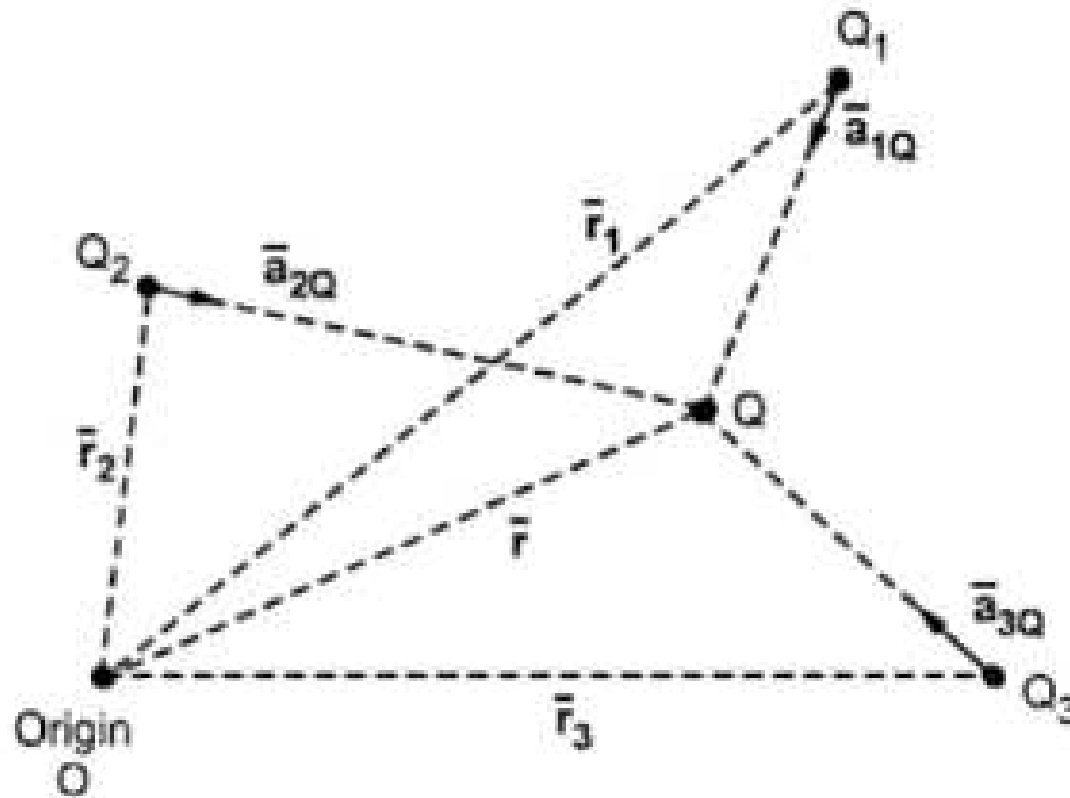
Vector form of Coulomb's law

$$\vec{F}_2 = \frac{Q_1 Q_2}{4\pi\epsilon_0 R_{12}^2} \vec{a}_{12}$$

$\vec{a}_{12}$ = Unit vector along $\vec{R}_{12} = \frac{\text{Vector}}{\text{Magnitude of vector}}$

$$\vec{a}_{12} = \frac{\vec{R}_{12}}{|\vec{R}_{12}|} = \frac{\vec{r}_2 - \vec{r}_1}{|\vec{R}_{12}|} = \frac{\vec{r}_2 - \vec{r}_1}{|\vec{r}_2 - \vec{r}_1|}$$

$|\vec{R}_{12}| = R = \text{Distance between the two charges}$



Hence the total force on Q is,

$$\vec{F}_Q = \vec{F}_{Q_1Q} + \vec{F}_{Q_2Q} + \vec{F}_{Q_3Q}$$

In general if there are n other charges then force exerted on Q due to all other n charges is,

$$\vec{F}_Q = \vec{F}_{Q_1Q} + \vec{F}_{Q_2Q} + \dots + \vec{F}_{Q_nQ}$$

$$\vec{F}_Q = \frac{Q}{4\pi\epsilon_0} \sum_{i=1}^n \frac{Q_i}{R_{iQ}^2} \frac{\vec{r} - \vec{r}_i}{|\vec{r} - \vec{r}_i|}$$

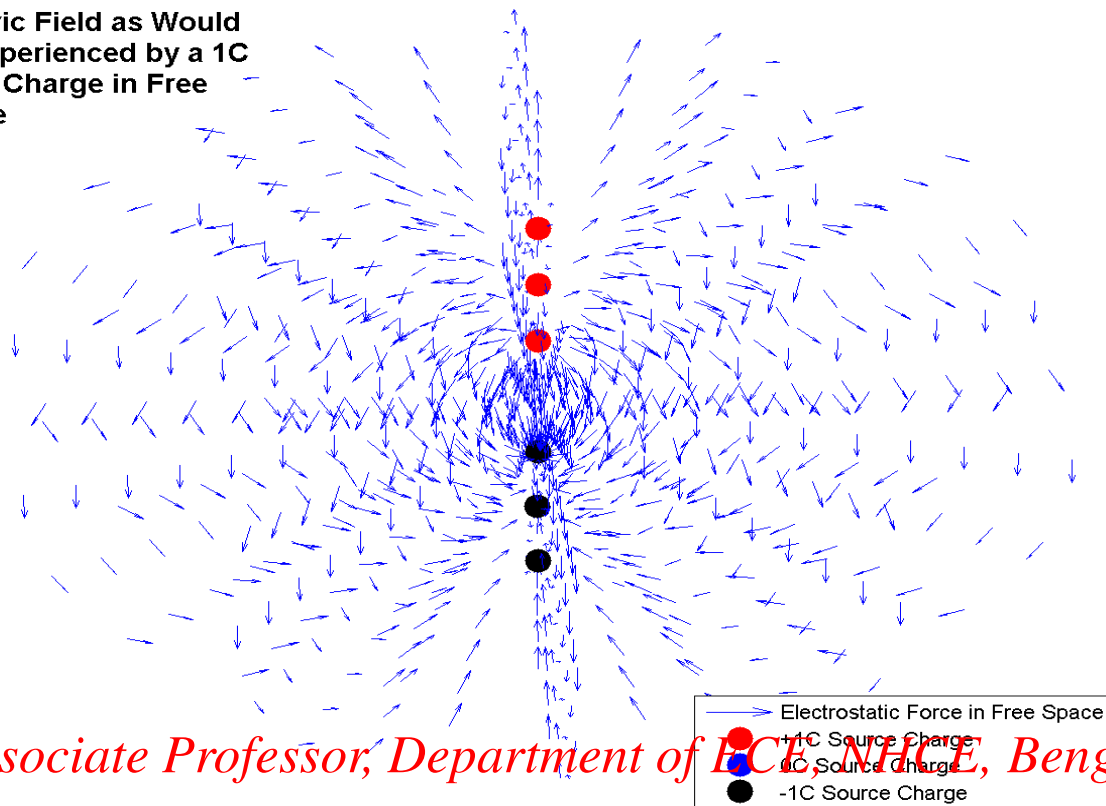
$\therefore$

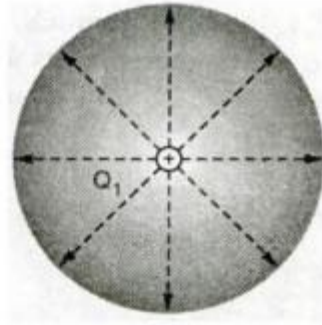
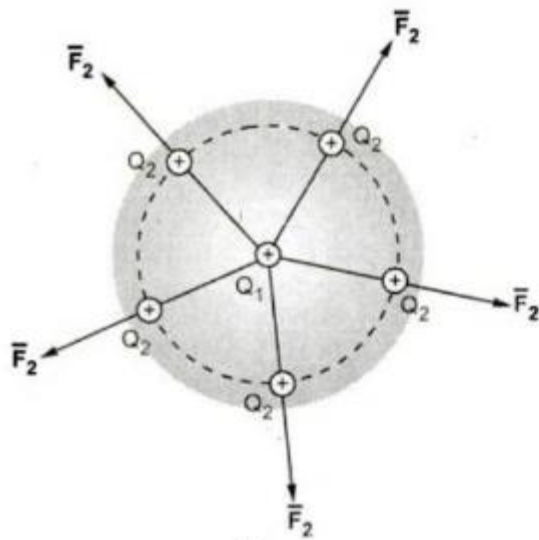
Electric Field Intensity:

Electric field strength or intensity is the force per unit charge when placed in the electric field.

It is also defined as the force experienced by a unit positive test charge.

Electric Field as Would
be Experienced by a 1C
Point Charge in Free
Space





Point charge Q_1 , Q_2 experiences force.

The force experienced by the charge Q_2 due to Q_1 is given by Coulomb's law as,

$$\vec{F}_2 = \frac{Q_1 Q_2}{4\pi\epsilon_0 R_{12}^2} \vec{a}_{12}$$

Thus force per unit charge can be written as,

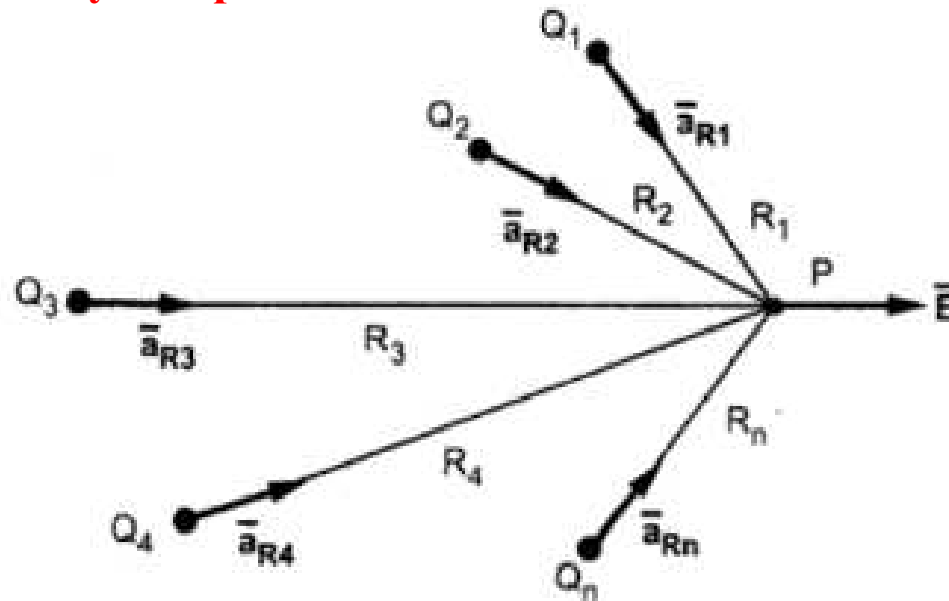
$$\frac{\vec{F}_2}{Q_2} = \frac{Q_1}{4\pi\epsilon_0 R_{12}^2} \vec{a}_{12}$$

$$\vec{E} = \frac{Q_1}{4\pi\epsilon_0 R_{1p}^2} \vec{a}_{1p}$$

$$\vec{E} = \frac{\text{Force}}{\text{Unit charge}} = \frac{(\text{N}) \text{ Newtons}}{(\text{C}) \text{ Coulomb}}$$

or V/m

Electric Field Intensity at a point due to 'n' number of Point charges:

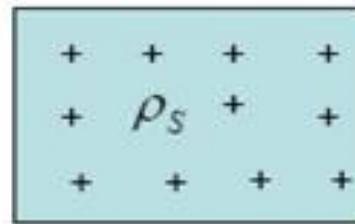
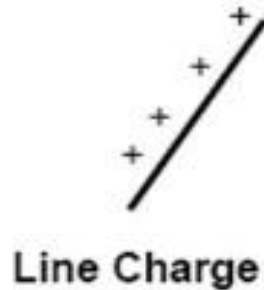
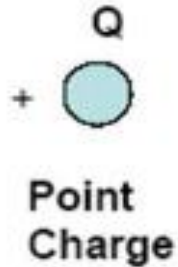


Then the total electric field intensity at point P is the vector sum of the individual field intensities produced by the various charges at the point P.

$$\begin{aligned}\vec{E} &= \vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \dots + \vec{E}_n \\ &= \frac{Q_1}{4\pi\epsilon_0 R_1^2} \vec{a}_{R1} + \frac{Q_2}{4\pi\epsilon_0 R_2^2} \vec{a}_{R2} + \dots + \frac{Q_n}{4\pi\epsilon_0 R_n^2} \vec{a}_{Rn}\end{aligned}$$

$$\boxed{\vec{E} = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{Q_i}{R_i^2} \vec{a}_{Ri}}$$

Types of charge Distributions:



Charge uniformly distributed in a line

* Finite or Infinite

Charge uniformly distributed in two dimensional surface

$$\rho_v = \frac{\text{Total charge in coulomb}}{\text{Total volume in cubic metres}} \left(\frac{\text{C}}{\text{m}^3} \right)$$

$$\rho_L = \frac{\text{Total charge in coulomb}}{\text{Total length in metres}} (\text{C/m})$$

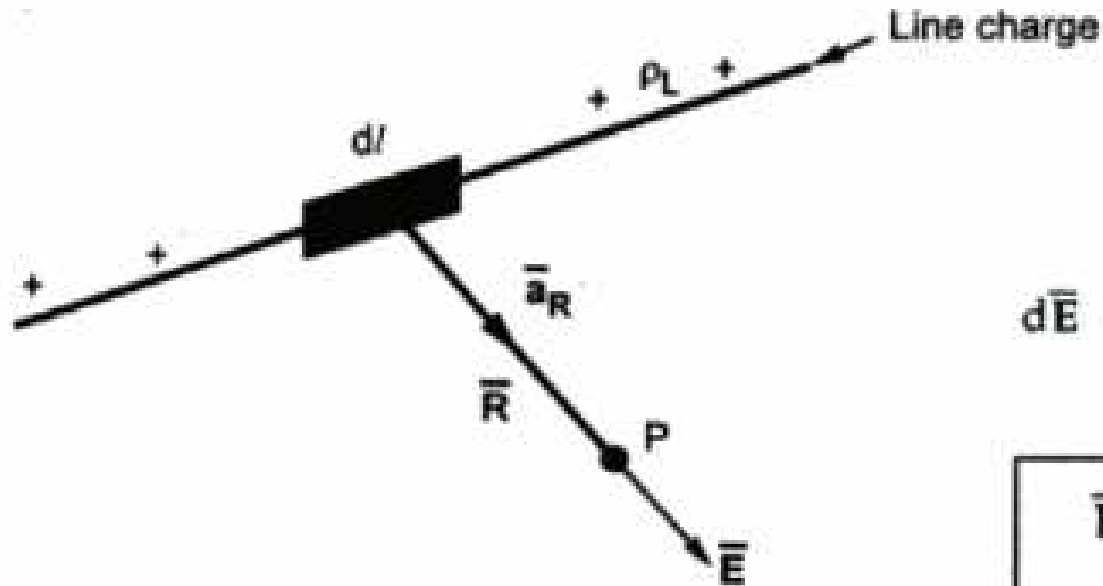
$$\rho_S = \frac{\text{Total charge in coulomb}}{\text{Total area in square metres}} (\text{C/m}^2)$$

$$Q = \int_{\text{vol}} \rho_v \, dv$$

$$Q = \oint_L \rho_L \, dl$$

$$Q = \int_S dQ = \int_S \rho_S \, dS$$

Electric Field Intensity due to finite line charge:



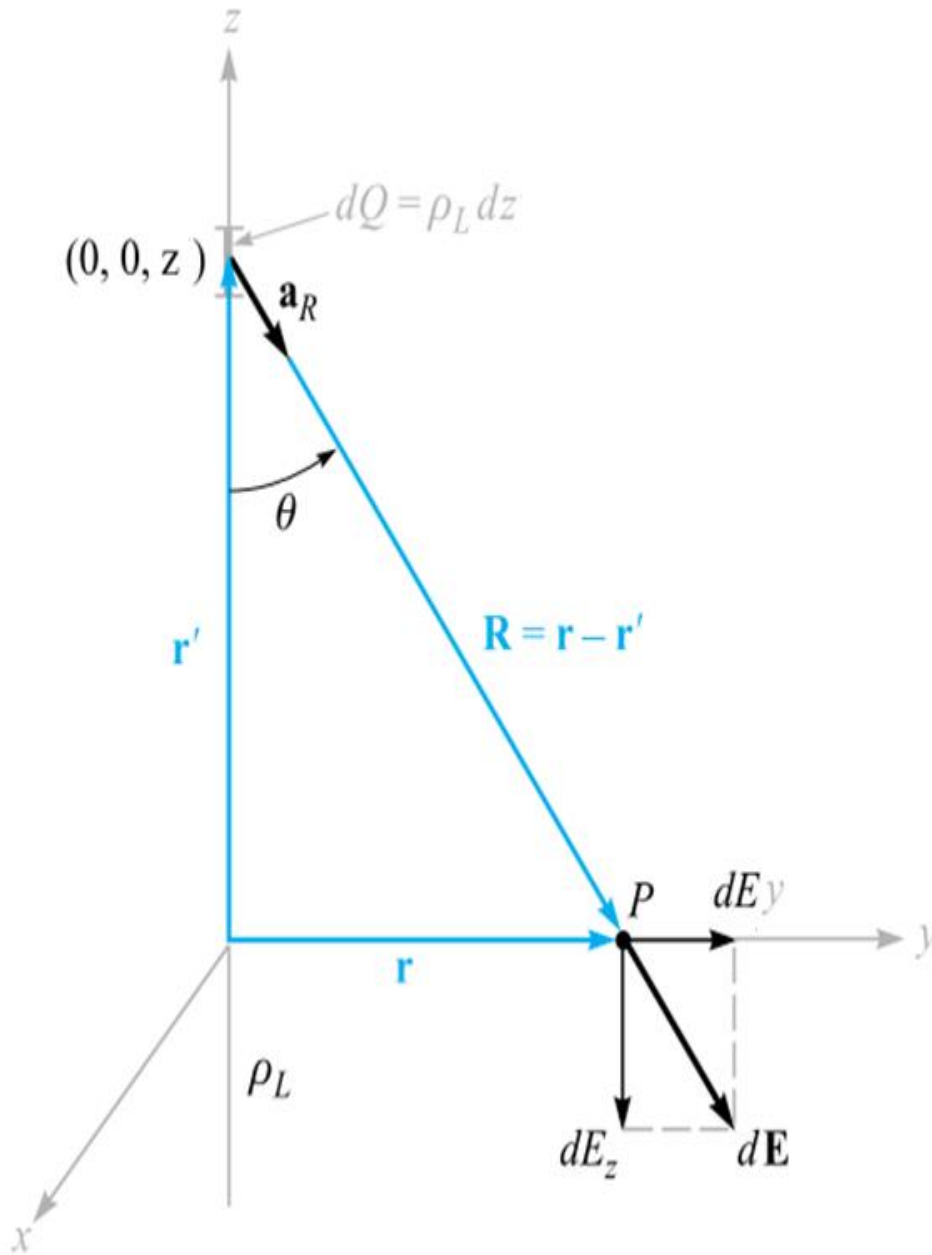
$$dQ = \rho_L dl$$

$$d\bar{E} = \frac{dQ}{4\pi\epsilon_0 R^2} \bar{a}_R = \frac{\rho_L dl}{4\pi\epsilon_0 R^2} \bar{a}_R$$

$$\bar{E} = \int_L \frac{\rho_L dl}{4\pi\epsilon_0 R^2} \bar{a}_R$$

$$\bar{E} = \frac{Q_1}{4\pi\epsilon_0 R_{ip}^2} \bar{a}_{ip}$$

Electric Field Intensity due to Infinite line charge:



Infinitely long straight line

Density is ρ_L

Line is along z-axis from ∞ to $-\infty$

Point 'P' - Electric field to be determined.

Distance from origin is 'r'

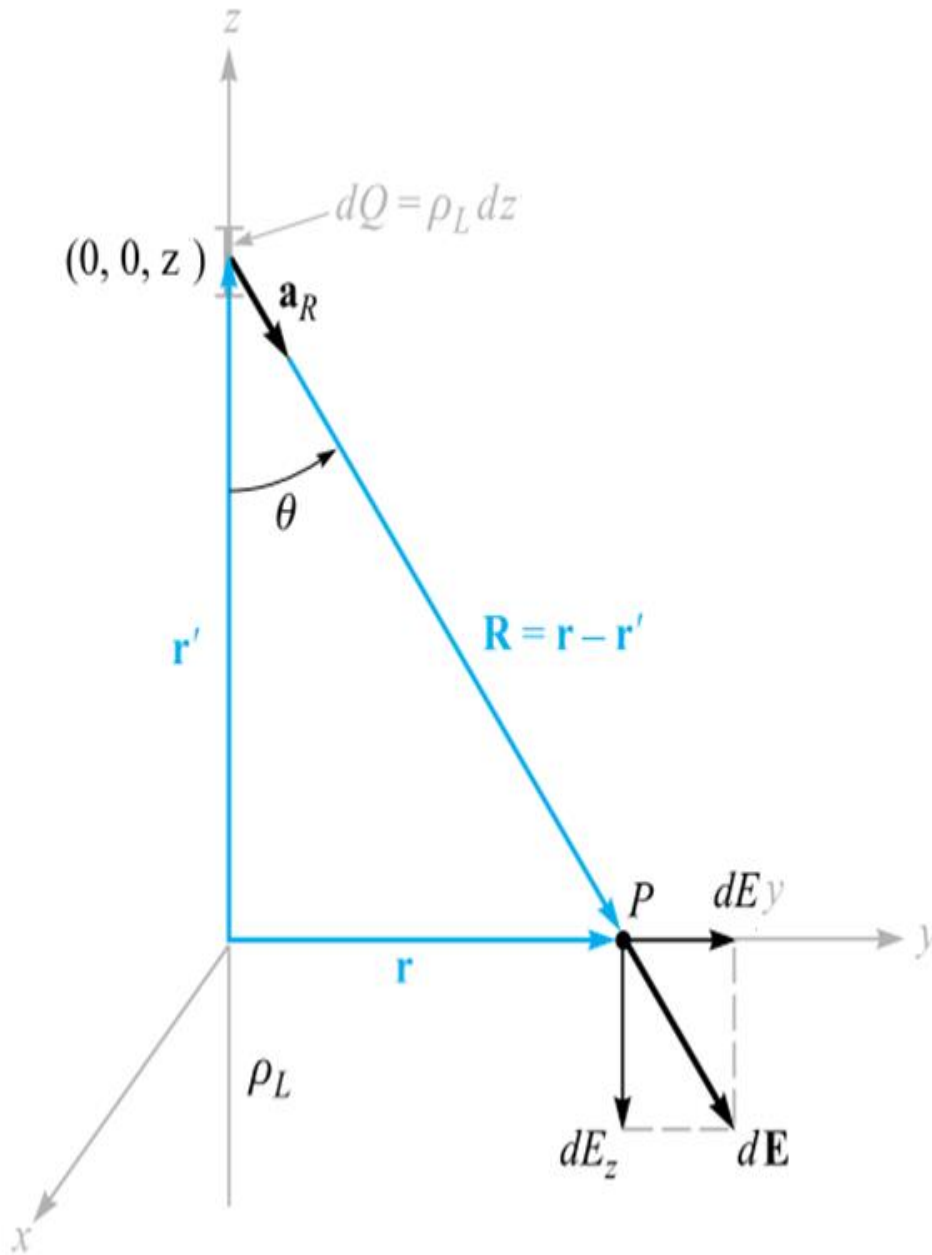
Consider small differential length dl carrying a charge dQ along the line.

$$dl = dz ;$$

$$dQ = \rho_L dl$$

$$= \rho_L dz$$

Electric Field Intensity due to Infinite line charge:



The coordinates of dQ are $(0,0,z)$ while the coordinates of point P are $(0,r,0)$.

Distance vector

$$\bar{\mathbf{R}} = \bar{\mathbf{r}}_P - \bar{\mathbf{r}}_{dQ} = [r \bar{\mathbf{a}}_y - z \bar{\mathbf{a}}_z]$$

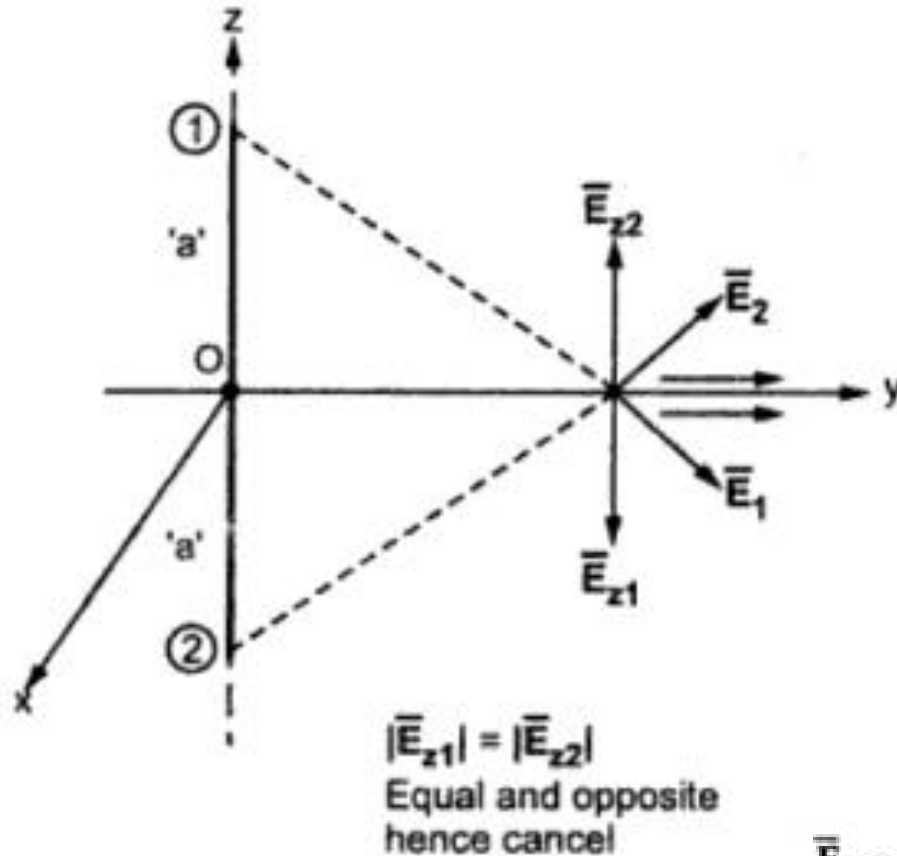
$$|\bar{\mathbf{R}}| = \sqrt{r^2 + z^2}$$

$$\bar{\mathbf{a}}_R = \frac{\bar{\mathbf{R}}}{|\bar{\mathbf{R}}|} = \frac{r \bar{\mathbf{a}}_y - z \bar{\mathbf{a}}_z}{\sqrt{r^2 + z^2}}$$

$$d\bar{\mathbf{E}} = \frac{dQ}{4\pi\epsilon_0 R^2} \bar{\mathbf{a}}_R$$

$$= \frac{\rho_L dz}{4\pi\epsilon_0 (\sqrt{r^2 + z^2})^2} \left[\frac{r \bar{\mathbf{a}}_y - z \bar{\mathbf{a}}_z}{\sqrt{r^2 + z^2}} \right]$$

Electric Field Intensity due to Infinite line charge:



$$d\vec{E} = \frac{\rho_L dz}{4\pi\epsilon_0 (\sqrt{r^2 + z^2})^2} \frac{r \vec{a}_y}{\sqrt{r^2 + z^2}}$$

$$\vec{E} = \int_{-\infty}^{\infty} \frac{\rho_L}{4\pi\epsilon_0 (r^2 + z^2)^{3/2}} r dz \vec{a}_y$$

$$\tan\theta = \frac{z}{r};$$

$$r = \frac{z}{\tan\theta};$$

$$dz = r \sec^2\theta d\theta$$

Hence r is not variable of integration

For $z = -\infty$,

$$\theta = \tan^{-1}(-\infty) = \frac{-\pi}{2} = -90^\circ$$

For $z = \infty$,

$$\theta = \tan^{-1}(\infty) = \frac{\pi}{2} = 90^\circ$$

$$\vec{E} = \int_{\theta=-\pi/2}^{\pi/2} \frac{\rho_L}{4\pi\epsilon_0 [r^2 + r^2 \tan^2\theta]^{3/2}} r \times r \sec^2\theta d\theta \vec{a}_y$$

$$= \frac{\rho_L}{4\pi\epsilon_0} \int_{-\pi/2}^{\pi/2} \frac{r^2 \sec^2\theta d\theta}{r^3 [1 + \tan^2\theta]^{3/2}} \vec{a}_y$$

$$1 + \tan^2\theta = \sec^2\theta$$

Electric Field Intensity due to Infinite line charge:

$$\begin{aligned}\bar{E} &= \int_{\theta=-\pi/2}^{\pi/2} \frac{\rho_L}{4\pi\epsilon_0 [r^2 + r^2 \tan^2 \theta]^{3/2}} r \times r \sec^2 \theta d\theta \bar{a}_y \\ &= \frac{\rho_L}{4\pi\epsilon_0} \int_{-\pi/2}^{\pi/2} \frac{r^2 \sec^2 \theta d\theta}{r^3 [1 + \tan^2 \theta]^{3/2}} \bar{a}_y\end{aligned}$$

$$1 + \tan^2 \theta = \sec^2 \theta$$

$$\begin{aligned}\bar{E} &= \frac{\rho_L}{4\pi\epsilon_0} \int_{-\pi/2}^{\pi/2} \frac{\sec^2 \theta d\theta}{r \sec^3 \theta} \bar{a}_y \\ &= \frac{\rho_L}{4\pi\epsilon_0 r} \int_{-\pi/2}^{\pi/2} \cos \theta d\theta \bar{a}_y \quad \dots \sec \theta = \frac{1}{\cos \theta}\end{aligned}$$

$$= \frac{\rho_L}{4\pi\epsilon_0 r} [\sin \theta]_{-\pi/2}^{\pi/2} \bar{a}_y = \frac{\rho_L}{4\pi\epsilon_0 r} \left[\sin \frac{\pi}{2} - \sin \left(-\frac{\pi}{2} \right) \right] \bar{a}_y$$

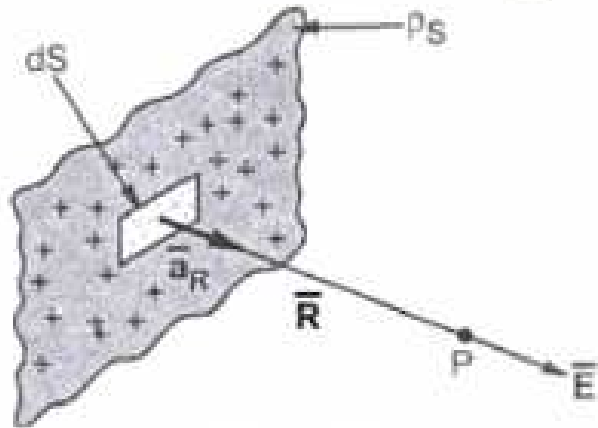
$$= \frac{\rho_L}{4\pi\epsilon_0 r} [1 - (-1)] \bar{a}_y = \frac{\rho_L}{4\pi\epsilon_0 r} \times 2 \bar{a}_y$$

$$\boxed{\bar{E} = \frac{\rho_L}{2\pi\epsilon_0 r} \bar{a}_y \text{ V/m}}$$

$$\vec{E} = \frac{\rho_L}{2\pi\epsilon_0 r} \vec{a}_r \text{ V/m}$$

Electric Field Intensity due to Surface Charge:

$\vec{E}$ due to Surface Charge



The charge dQ on the differential surface area dS is,

$$dQ = \rho_s dS$$

Hence the differential electric field $d\vec{E}$ at a point P due to dQ is given by,

$$d\vec{E} = \frac{dQ}{4\pi\epsilon_0 R^2} \vec{a}_R = \frac{\rho_s dS}{4\pi\epsilon_0 R^2} \vec{a}_R$$

Hence the total $\vec{E}$ at a point P is to be obtained by integrating $d\vec{E}$ over the surface area on which charge is distributed. Note that this will be a double integration.

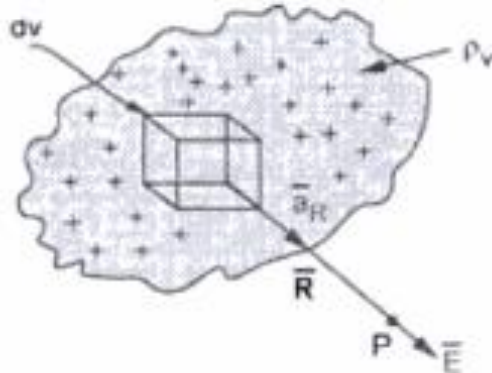
$\therefore$

$$\vec{E} = \int_S \frac{\rho_s dS}{4\pi\epsilon_0 R^2} \vec{a}_R$$

The $\vec{a}_R$ and dS to be obtained according to the position of the sheet of charge and the co-ordinate system used.

Electric Field Intensity due to Volume Charge:

$\vec{E}$ due to Volume Charge



The charge dQ on the differential volume dv is,

$$dQ = \rho_v dv$$

Hence the differential electric field $d\vec{E}$ at a point P due to dQ is given by,

$$d\vec{E} = \frac{dQ}{4\pi\epsilon_0 R^2} \vec{a}_R = \frac{\rho_v dv}{4\pi\epsilon_0 R^2} \vec{a}_R$$

Hence the total $\vec{E}$ at a point P is to be obtained by integrating $d\vec{E}$ over the volume in which charge is accumulated. Note that this integration will be a triple integration.

$\therefore$

$$\vec{E} = \int_{Vol} \frac{\rho_v dv}{4\pi\epsilon_0 R^2} \vec{a}_R$$

The $\vec{a}_R$ and dv must be obtained according to the co-ordinate system used.

Summary of Electric Field Intensity due to Various charge distributions

$$\vec{E}_{\text{total}} = \vec{E}_p + \vec{E}_l + \vec{E}_s + \vec{E}_v$$

where $\vec{E}_p$, $\vec{E}_l$, $\vec{E}_s$ and $\vec{E}_v$ are the field intensities due to point, line, surface and volume charge distributions respectively.

$$\vec{E} = \frac{Q_1}{4\pi\epsilon_0 R_{ip}^2} \vec{a}_{ip}$$

Point charge

Line charge

$$\vec{E} = \int_L \frac{\rho_L dl}{4\pi\epsilon_0 R^2} \vec{a}_R$$

Infinite Line charge

$$\vec{E} = \frac{\rho_L}{2\pi\epsilon_0 r} \vec{a}_r \text{ V/m}$$

$$\vec{E} = \int_S \frac{\rho_S dS}{4\pi\epsilon_0 R^2} \vec{a}_R$$

Surface charge

$$\vec{E} = \int_{\text{Vol}} \frac{\rho_v dv}{4\pi\epsilon_0 R^2} \vec{a}_R$$

Volume charge

Electric flux density, Gauss Law and Divergence:

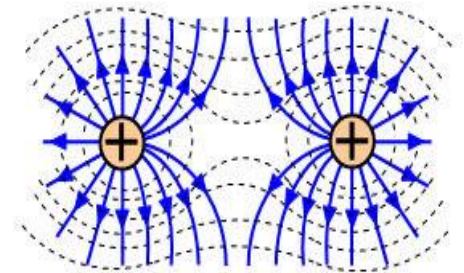
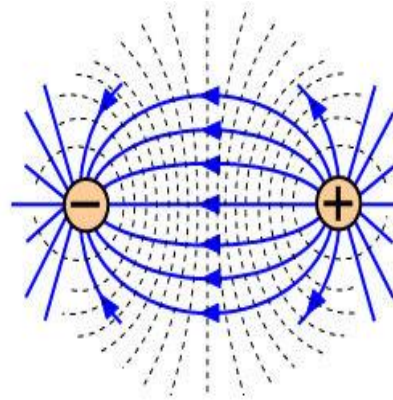
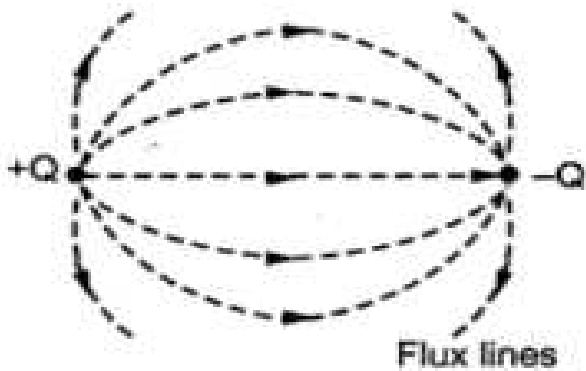
Flux - the action or process of flowing or flowing out.

Electric flux – the electric lines of force radially coming out from positive charge.

Displacement Flux

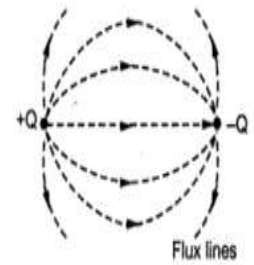
If a unit test charge is placed near a point charge, it experiences a force.

The direction of this force can be represented by the lines, radially coming outward from a positive charge. These lines are called streamlines or flux lines.



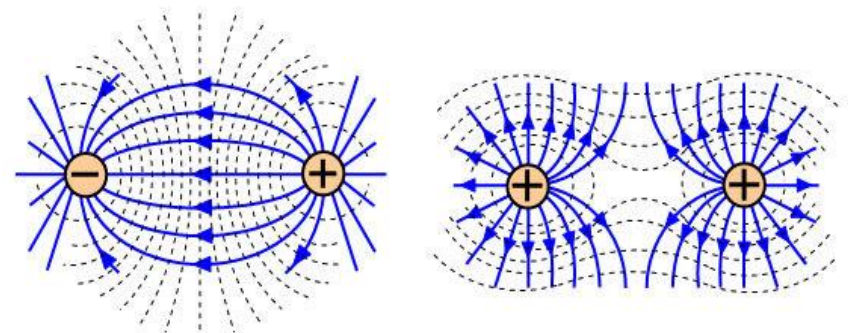
Unit – coulomb, Symbol - Ψ

Electric flux /displacement flux



Properties of flux lines:

1. Start from positive – terminate at negative
2. Negative charge is absent – flux lines terminate at infinity
3. Electric field is stronger/strength/intensity is high ---- Lines are more
4. Lines are parallel
5. Independent of the medium in which charges are placed
6. Lines always enter or leave the charged surface normally.
7. Electric flux = $Q = \Psi$ (numerically) The total number of lines = charge on the body

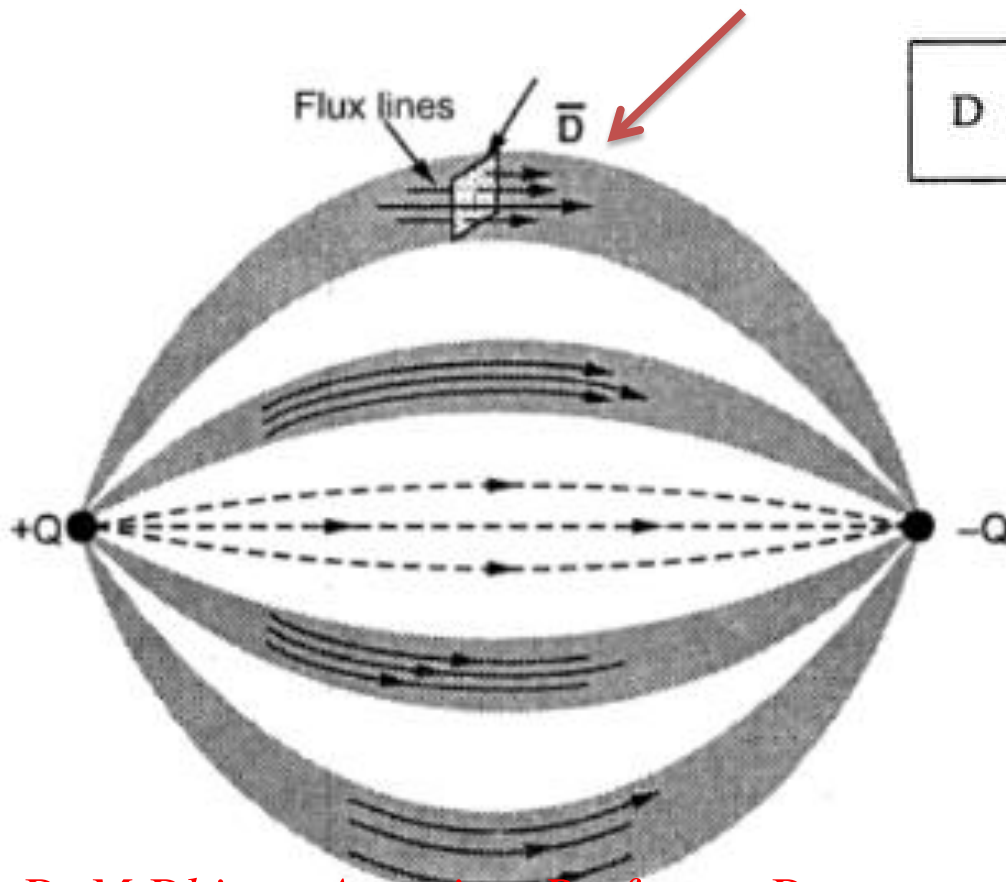


Electric flux density (D)

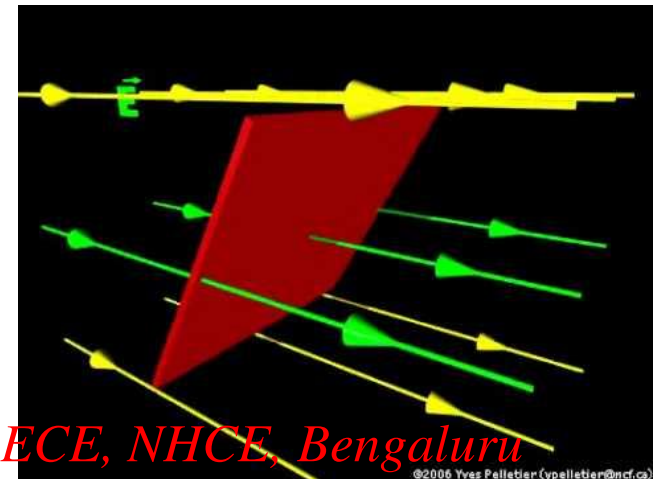
Flux – scalar quantity

Electric flux density / Displacement Flux density– Vector quantity

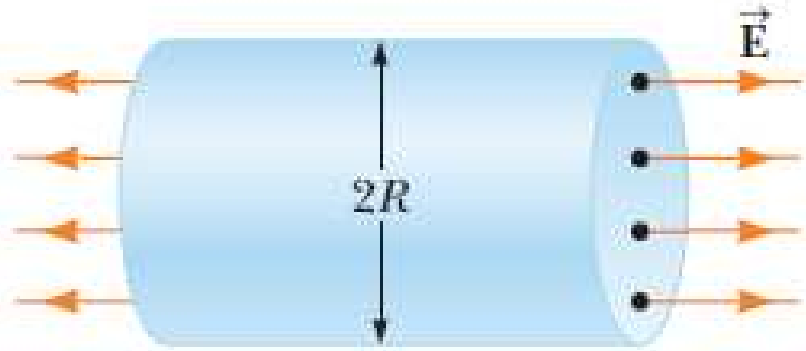
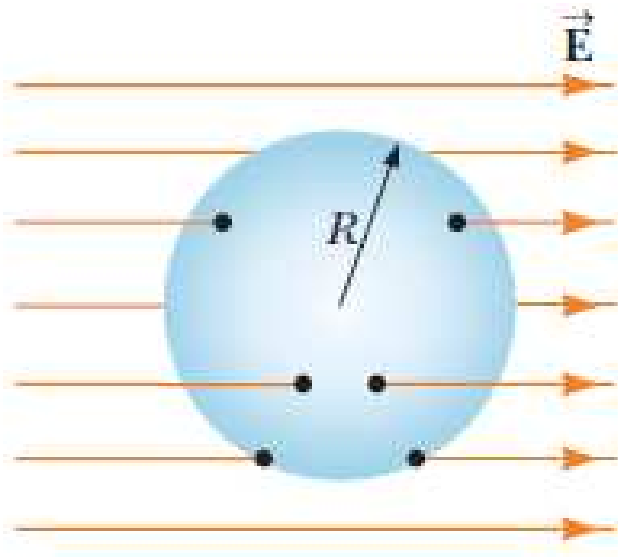
Net flux passing normal through the unit surface area



$$D = \frac{\Psi}{S} \text{ in magnitude } \text{C/m}^2.$$



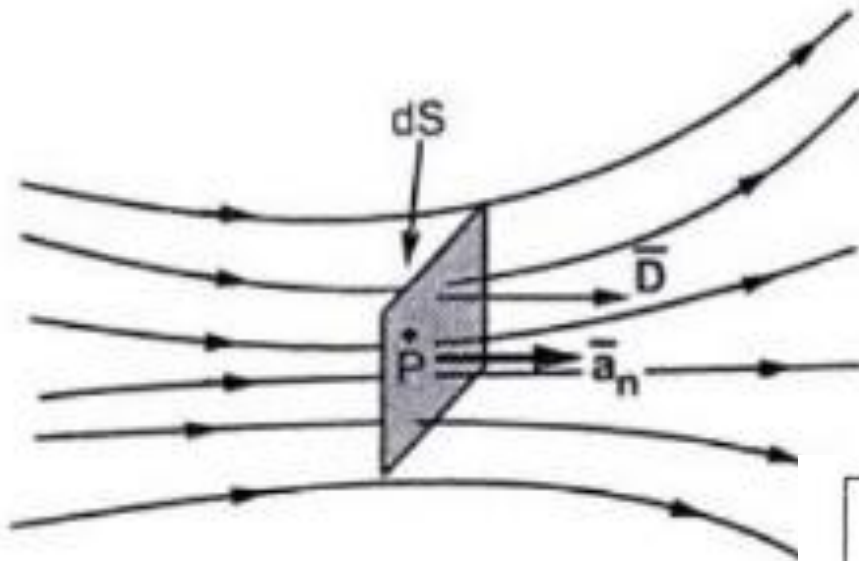
Dr.M.Dhivya, Associate Professor, Department of ECE, NHCE, Bengaluru



A closed surface could be the surface area of a sphere or a cube, etc. Flux is measured at a single point in time. The total **flux depends on** strength of the field, the size of the surface it passes through, and their orientation.

- a point charge
- a uniformly distributed spherical shell of charge
- any other charge distribution with spherical symmetry

- an infinitely long line of uniform charge
- an infinite plane of uniform charge
- an infinitely long cylinder of uniform charge



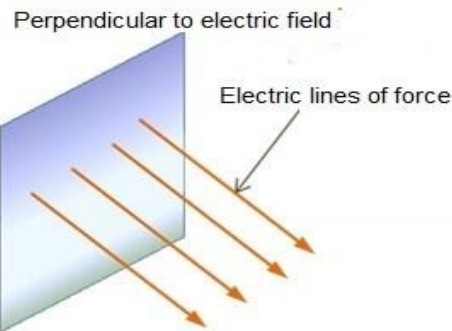
Flux through dS

$$\bar{D} = \frac{d\psi}{dS} \bar{a}_n \text{ C/m}^2$$

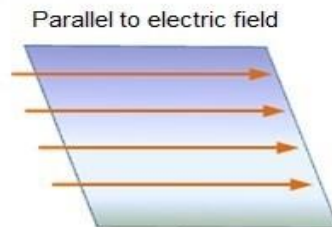
$d\psi$ = Total flux lines crossing normal through the differential area dS

dS = Differential surface area

$\bar{a}_n$ = Unit vector in the direction normal to the differential surface area



Maximum electric flux

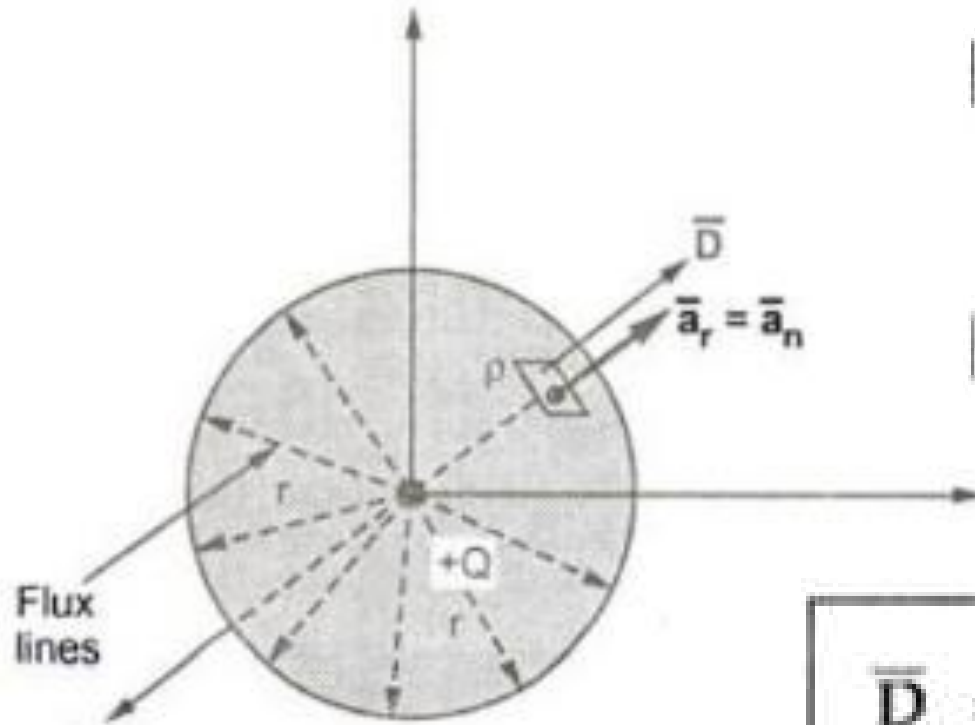


Zero electric flux

- Differential surface area ds at point p .
- Differential flux is $d\psi$.
- Differential area and flux are at right angles.
- D vector is normal to the surface area.

If the electric field is lying ***along*** the surface, it isn't going in or going out and the flux is zero.

Electric Flux Density due to point Charge Q



$$|\bar{D}| = \frac{\text{Total flux } \psi}{\text{Total surface area } S}$$

$$\psi = Q = \text{Total flux}$$

$$S = 4\pi r^2 = \text{Total surface area}$$

$$|\bar{D}| = \frac{Q}{4\pi r^2}$$

$$\bar{D} = \frac{Q}{4\pi r^2} \bar{a}_r \text{ C/m}^2$$

Relationship between D and E

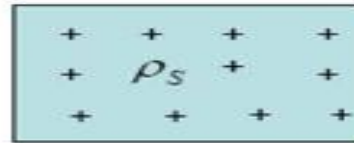
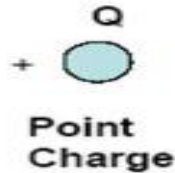
$$\bar{E} = \frac{Q}{4\pi\epsilon_0 r^2} \bar{a}_r$$

$$\frac{\bar{D}}{\bar{E}} = \frac{\frac{Q}{4\pi r^2} \bar{a}_r}{\frac{Q}{4\pi\epsilon_0 r^2} \bar{a}_r} = \epsilon_0$$

$$\bar{D} = \epsilon_0 \bar{E}$$

$$\bar{D} = \epsilon \bar{E}$$

Types of charge Distributions:



$$\rho_L = \frac{\text{Total charge in coulomb}}{\text{Total length in metres}} \text{ (C/m)}$$

$$Q = \int_L \rho_L dl$$

$$\vec{D} = \frac{Q}{4\pi r^2} \vec{a}_r = \frac{\int_L \rho_L dl}{4\pi r^2} \vec{a}_r$$

$$\vec{E} = \frac{\rho_L}{2\pi\epsilon_0 r} \vec{a}_r$$

$$\vec{D} = \epsilon_0 \vec{E}$$

$$\vec{D} = \frac{\rho_L}{2\pi r} \vec{a}_r$$

$$\rho_S = \frac{\text{Total charge in coulomb}}{\text{Total area in square metres}} \text{ (C/m}^2\text{)}$$

$$Q = \int_S dQ = \int_S \rho_S dS$$

$$\vec{D} = \frac{Q}{4\pi r^2} \vec{a}_r = \frac{\int_S \rho_S dS}{4\pi r^2} \vec{a}_r$$

$$\vec{E} = \frac{\rho_S}{2\epsilon_0} \vec{a}_n$$

$$\vec{D} = \epsilon_0 \vec{E}$$

$$\vec{D} = \frac{\rho_S}{2} \vec{a}_n$$

$$\rho_V = \frac{\text{Total charge in coulomb}}{\text{Total volume in cubic metres}} \left(\frac{\text{C}}{\text{m}^3} \right)$$

$$Q = \int_{\text{vol}} \rho_V dv$$

$$\vec{E} = \frac{\int_{\text{vol}} \rho_V dv}{4\pi\epsilon_0 r^2} \vec{a}_r$$

$$\vec{D} = \epsilon_0 \vec{E}$$

$$\vec{D} = \frac{\int_{\text{vol}} \rho_V dv}{4\pi r^2} \vec{a}_r$$

Gauss Law:

Sphere --- irregular closed surface --- Faraday

Type of distribution	----- Point Charge, Line charge, surface charge
Type of surface	----- cylinder, cube, sphere, Parabolic elliptic, irregular surface
etc....	

Total electric flux passing through the closed surface = Total Charge enclosed by the surface.

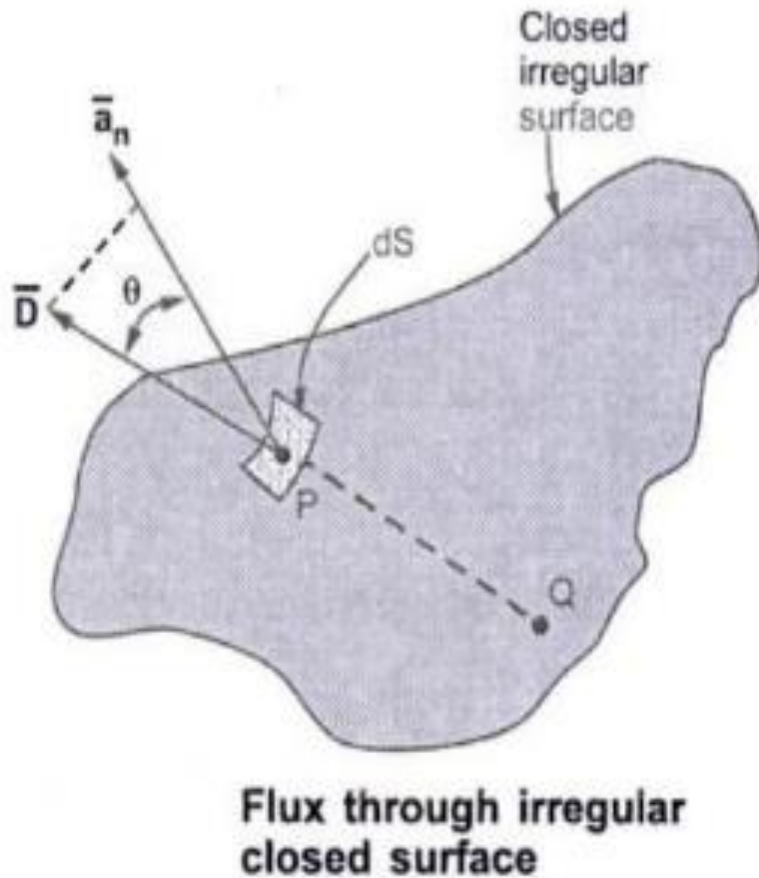
Applications: Used to determine electric field Intensity (E), Electric flux density (D)

Limitations: Gauss Law cannot be applied for asymmetrical/ disproportionate charge distribution.

Gauss Law:

Statement of Gauss's law :

The electric flux passing through any closed surface is equal to the total charge enclosed by that surface.

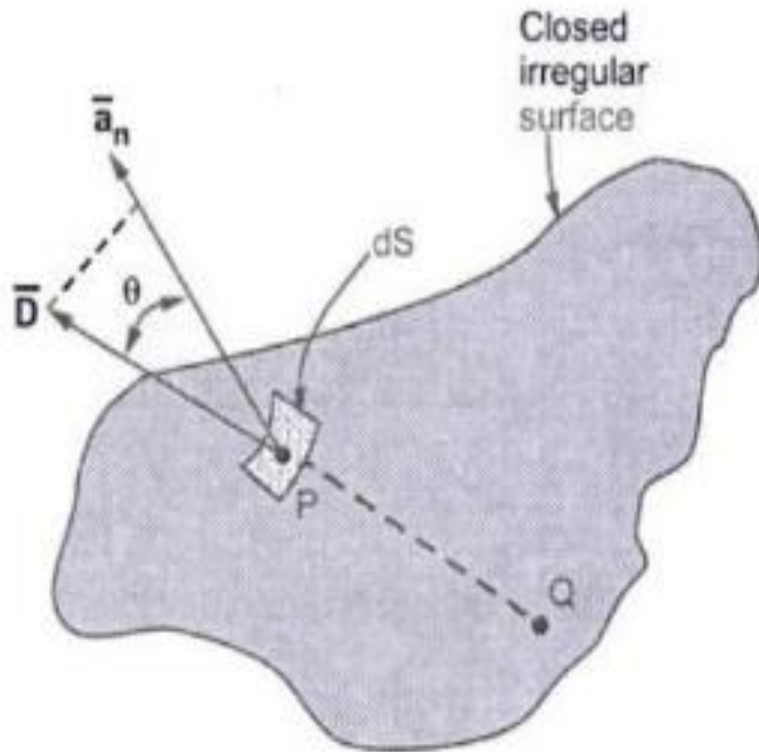


- Total charge enclosed = Q
- Any distribution
- Total flux that has to pass through the closed surface = Q .
- ds -small differential surface
- Direction of D will change from point to point on the surface.

Gauss Law:

Statement of Gauss's law :

The electric flux passing through any closed surface is equal to the total charge enclosed by that surface.



Flux through irregular closed surface

$$d\vec{S} = dS \vec{a}_n$$

$\vec{a}_n$ = Normal to the surface dS at point P

- Flux density D is making an angle θ with the normal direction at point P .

$$d\psi = D_n dS$$

D_n = Component of $\vec{D}$ in the direction of normal to the surface dS

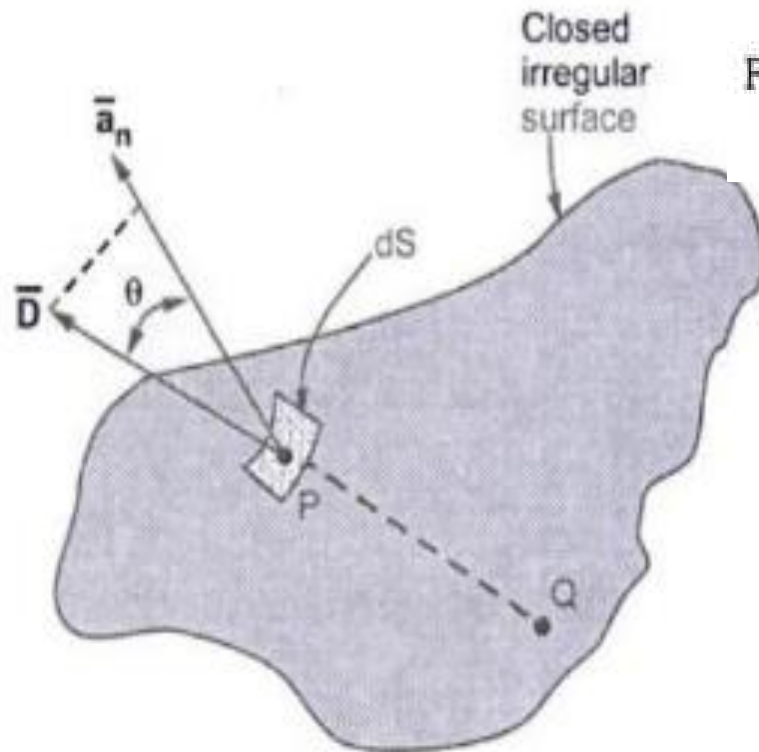
$$D_n = |\vec{D}| \cos\theta$$

$$d\psi = |\vec{D}| \cos\theta dS$$

Gauss Law:

Statement of Gauss's law :

The electric flux passing through any closed surface is equal to the total charge enclosed by that surface.



Flux through irregular closed surface

From the definition of the dot product, $\vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos \theta_{AB}$

$$|\vec{D}| dS \cos \theta = \vec{D} \cdot d\vec{S}$$

$$d\psi = \vec{D} \cdot d\vec{S}$$

$$\psi = \int d\psi = \oint_S \vec{D} \cdot d\vec{S}$$

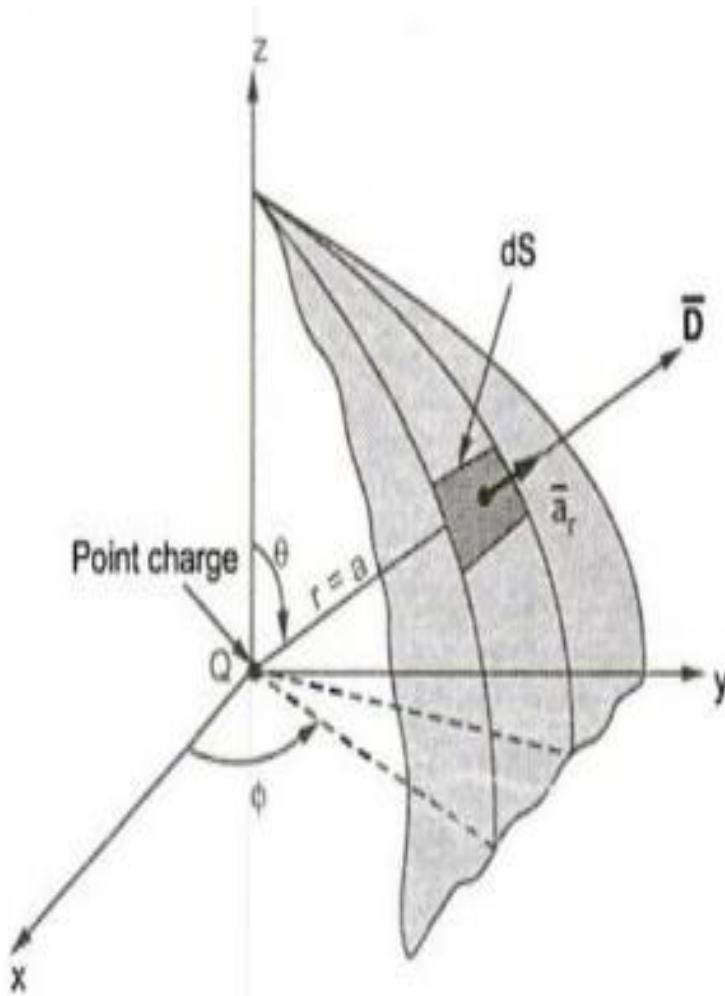
$$\psi = \oint_S \vec{D} \cdot d\vec{S} = Q = \text{Charge enclosed}$$

Gauss Law to various charge distributions:

- 1. Point charge**
- 2. Infinite line Charge**
- 3. Infinite Sheet of Charge**
- 4. Spherical Shell of Charge**
- 5. Uniformly Charged Sphere**
- 6. Gauss Law applied to differential volume element**

Gauss Law to various charge distributions:

1. Determination of field due to Point charge



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Gauss law is given by

$$\int_S \vec{D} \cdot \vec{ds} = Q$$

$$\vec{D} = D_r \cdot \vec{a}_r \quad ds = r^2 \sin\theta d\theta d\phi \vec{a}_r$$

$$\int_S D_r \cdot r^2 \sin\theta d\theta d\phi = Q$$

$$\int_0^\pi \int_0^{2\pi} D_r \cdot r^2 \sin\theta d\theta d\phi = D_r \cdot r^2 \int_0^\pi \int_0^{2\pi} \sin\theta d\theta d\phi$$

$$= D_r r^2 \int_0^\pi 2\pi \sin\theta d\theta = 2\pi D_r r^2 [-\cos\theta]_0^\pi$$

$$= 2\pi D_r r^2 [1+1] = 4\pi D_r r^2$$

$$D_r = \frac{Q}{4\pi r^2}$$

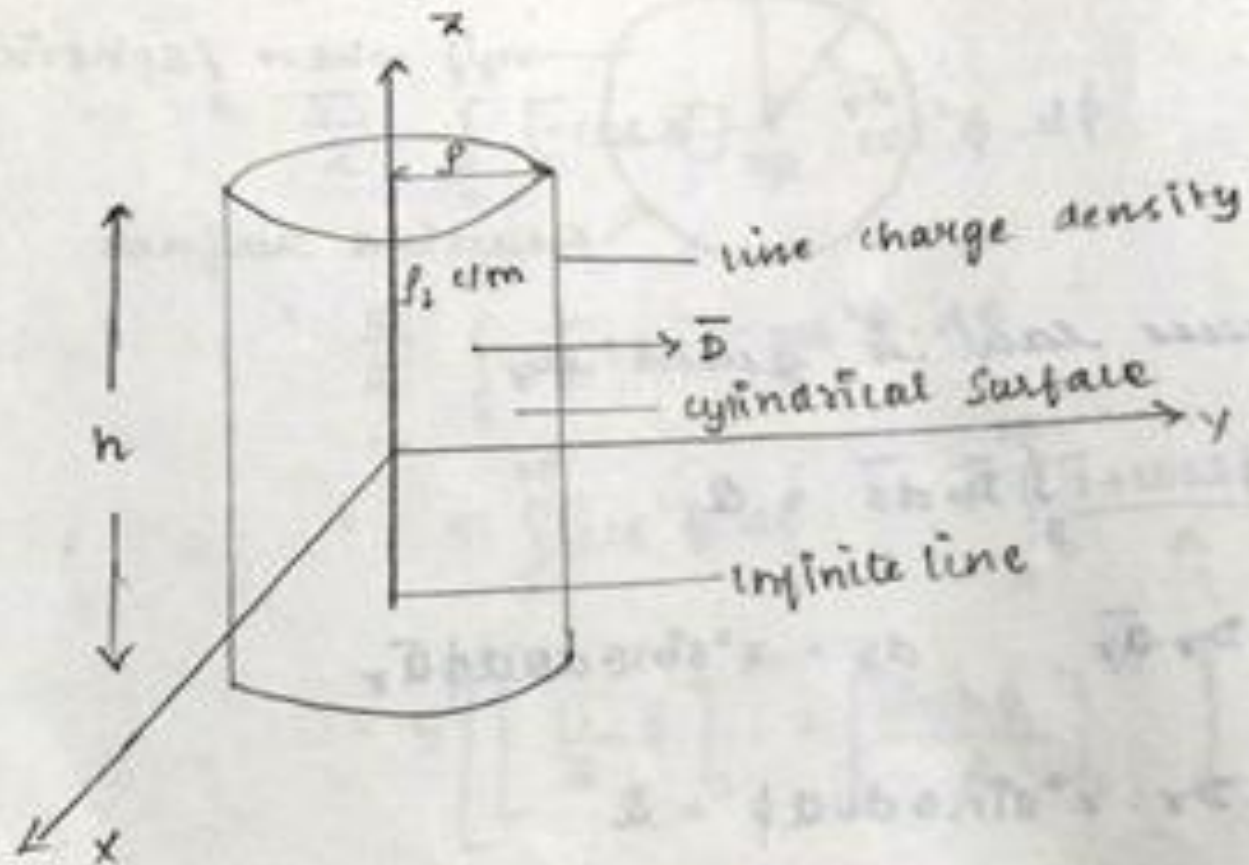
$$\bar{D} = D_r \cdot \bar{a}_r = \frac{Q}{4\pi r^2} \cdot \bar{a}_r$$

$$\bar{D} = \epsilon_0 \bar{E}$$

$$\bar{E} = \frac{\bar{D}}{\epsilon_0}$$

$$\bar{E} = \frac{Q}{4\pi \epsilon_0 r^2} \cdot \bar{a}_r$$

(2) Determination of field due to infinite line charge



Gauss law is given as

$$\oint_S \vec{D} \cdot d\vec{s} = Q$$

$$\bar{D} = D_f \cdot \bar{a}_f$$

$$ds = f d\phi dz \bar{a}_f$$

$$\begin{aligned} \int_s D_f \cdot f d\phi dz &= \int_0^h \int_0^{2\pi} D_f \cdot f d\phi dz \\ &= \int_0^h D_f \cdot f [\phi]_0^{2\pi} dz \\ &= D_f \cdot f \cdot 2\pi \int_0^h dz \\ &= 2\pi D_f \cdot f \cdot [z]_0^h \\ &= 2\pi h D_f \cdot f \end{aligned}$$

$$2\pi \cancel{D_f} \cdot f = Q = f_L \cdot \cancel{h}$$

$$D_f = \frac{f_L}{2\pi f}$$

$$\bar{D} = \frac{f_L}{2\pi f} \cdot \bar{a}_f$$

$$\bar{D} = \epsilon_0 \bar{E}$$

$$\bar{E} = \frac{\bar{D}}{\epsilon_0}$$

$$\bar{E} = \frac{f_L}{2\pi \epsilon_0 f} \cdot \bar{a}_f$$

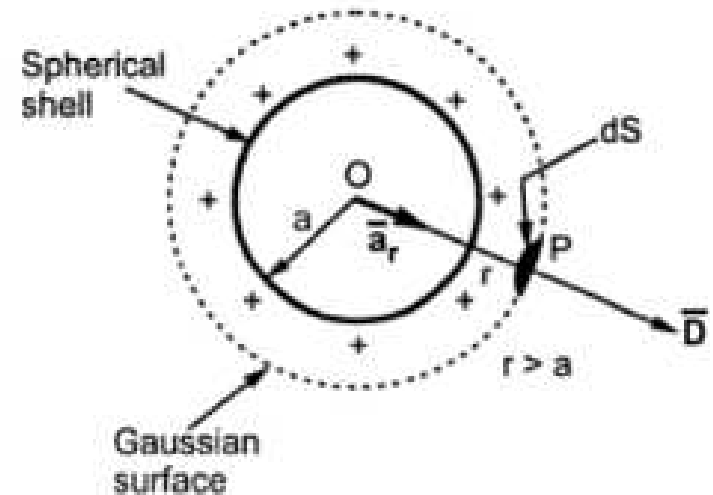
Gauss Law to various charge distributions:

1. Spherical Shell of Charge

Case i) Point P outside Shell $r > a$

Case ii) Point P is on the shell $r = a$

Case iii) Point P inside the shell $r < a$



Gauss Law to various charge distributions:

1. Spherical Shell of Charge

Case i) Point P outside Shell $r > a$

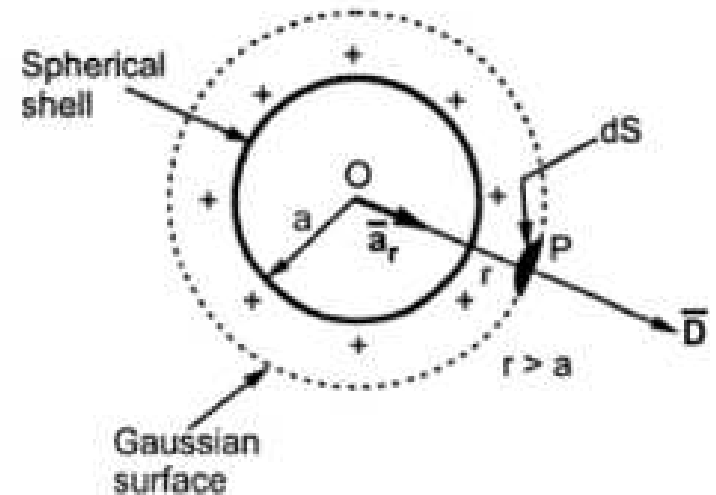
$$\begin{aligned} d\psi &= \vec{D} \cdot d\vec{S} = [D_r \vec{a}_r] \cdot [r^2 \sin\theta \, d\theta \, d\phi \vec{a}_r] \\ &= D_r r^2 \sin\theta \, d\theta \, d\phi \end{aligned}$$

$$\psi = \oint_S D_r r^2 \sin\theta \, d\theta \, d\phi = D_r r^2 \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} \sin\theta \, d\theta$$

$$\psi = D_r r^2 [-\cos\theta]_0^{\pi} [\phi]_0^{2\pi} = 4\pi r^2 D_r$$

$$\psi = Q$$

$$Q = 4\pi r^2 D_r$$



$$D_r = \frac{Q}{4\pi r^2}$$

$$\vec{D} = D_r \vec{a}_r = \frac{Q}{4\pi r^2} \vec{a}_r \text{ C/m}^2$$

$$\vec{E} = \frac{\vec{D}}{\epsilon_0} = \frac{Q}{4\pi\epsilon_0 r^2} \vec{a}_r \text{ V/m}$$

Gauss Law to various charge distributions:

1. Spherical Shell of Charge

Case i) Point P outside Shell $r > a$

If the surface charge density is ρ_s C/m² then

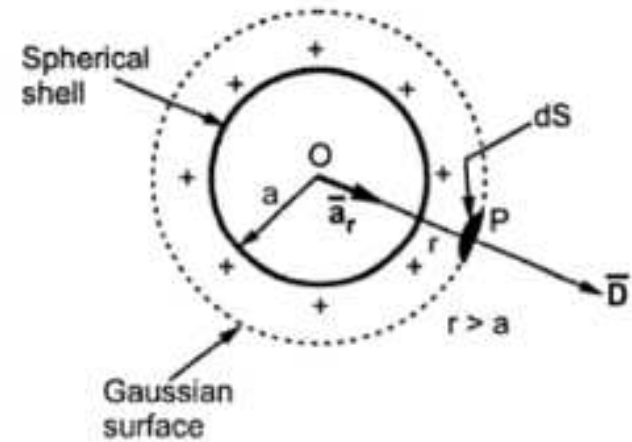
$$Q = \rho_s \times \text{Surface area of shell}$$

$$\therefore Q = \rho_s \times 4\pi a^2$$

$$\therefore \bar{E} = \frac{\rho_s 4\pi a^2}{4\pi\epsilon_0 r^2} \bar{a}_r = \frac{\rho_s a^2}{\epsilon_0 r^2} \bar{a}_r \text{ V/m}$$

and

$$\bar{D} = \epsilon_0 \bar{E} = \frac{\rho_s a^2}{r^2} \bar{a}_r \text{ C/m}^2$$



$$\bar{E} = \frac{Q}{4\pi\epsilon_0 a^2} \bar{a}_r \text{ V/m}$$

Case ii) Point P is on the shell, $r=a$.

Gauss Law to various charge distributions:

Case iii) Point P inside the shell $r < a$

$$\psi = Q = \oint_S \vec{D} \cdot d\vec{S} = 0 \quad \dots \text{As per Gauss's law}$$

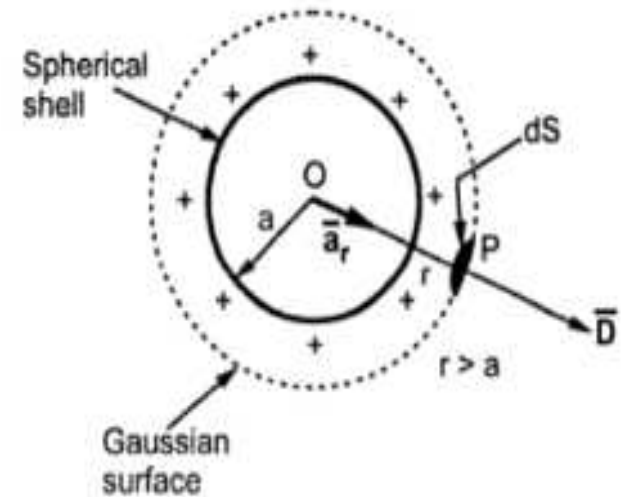
$$\oint_S d\vec{S} = \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} r^2 \sin\theta \, d\theta \, d\phi = 4\pi r^2$$

$$\oint_S d\vec{S} \neq 0$$

Hence to satisfy that total charge enclosed is zero, inside the spherical shell.

$$\vec{D} = 0 \quad \text{and} \quad \vec{E} = \frac{\vec{D}}{\epsilon_0} = 0$$

Thus electric flux density and electric field at any point inside a spherical shell is **zero**.



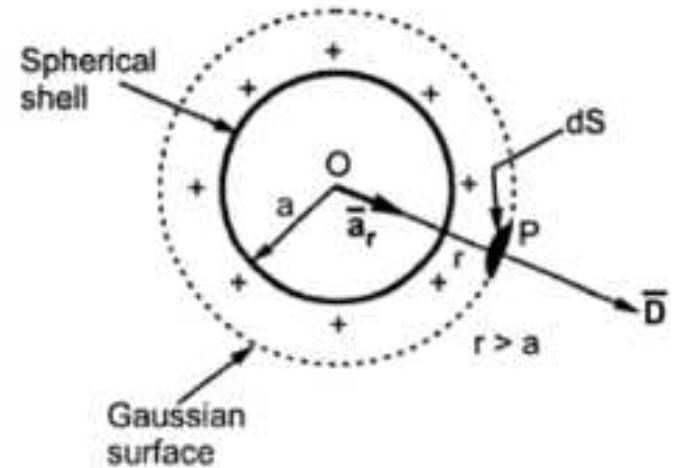
Gauss Law to various charge distributions:

Spherical Shell of Charge

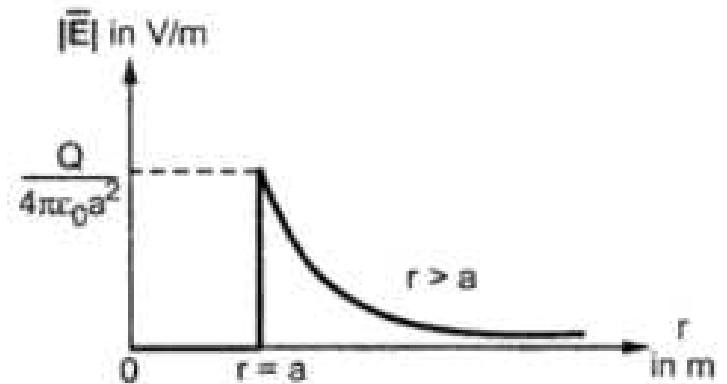
Case i) Point P outside Shell $r > a$

Case ii) Point P is on the shell $r = a$

Case iii) Point P inside the shell $r < a$



For $r < a$,	$\vec{E} = 0$
For $r = a$,	$\vec{E} = \frac{Q}{4\pi\epsilon_0 a^2} \vec{a}_r$
For $r > a$,	$\vec{E} = \frac{Q}{4\pi\epsilon_0 r^2} \vec{a}_r$



Variation of $|\vec{E}|$ against r

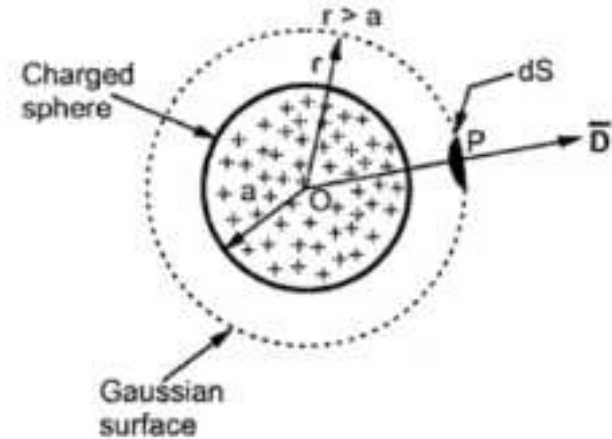
Gauss Law to various charge distributions:

Uniformly charged sphere

Case i) Point P outside sphere $r > a$

Case ii) Point P is on the Sphere $r = a$

Case iii) Point P inside the Sphere $r < a$



$$D_r = \frac{Q}{4\pi r^2}$$

$$\vec{D} = D_r \vec{a}_r = \frac{Q}{4\pi r^2} \vec{a}_r \text{ C/m}^2$$

$$\vec{E} = \frac{\vec{D}}{\epsilon_0} = \frac{Q}{4\pi\epsilon_0 r^2} \vec{a}_r \text{ V/m}$$

Gauss Law to various charge distributions:

The total charge enclosed can be obtained as,

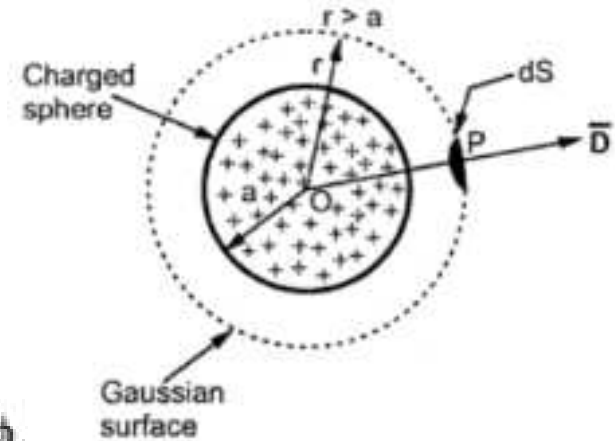
$$Q = \int_V \rho_v dv = \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} \int_{r=0}^a \rho_v r^2 \sin\theta dr d\theta d\phi$$

$$= \rho_v \left[\frac{r^3}{3} \right]_0^a [-\cos\theta]_0^{\pi} [\phi]_0^{2\pi}$$

$$= \frac{4}{3} \pi a^3 \rho_v \text{ C}$$

$$\vec{E} = \frac{\frac{4}{3} \pi a^3 \rho_v}{4 \pi \epsilon_0 r^2} \vec{a}_r = \frac{a^3 \rho_v}{3 \epsilon_0 r^2} \vec{a}_r$$

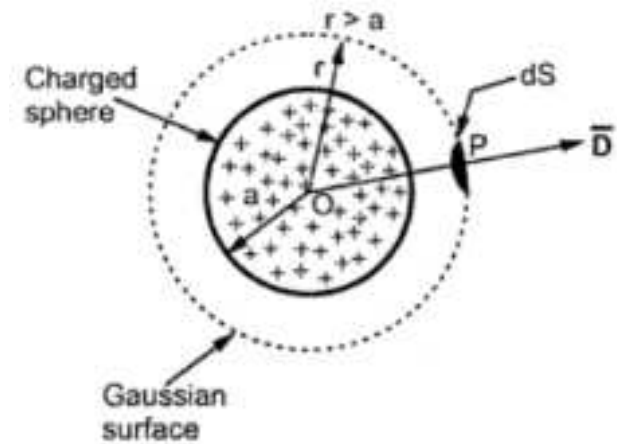
$$\boxed{\vec{D} = \frac{a^3 \rho_v}{3 r^2} \vec{a}_r}$$



Gauss Law to various charge distributions:

Uniformly charged sphere

Case ii) Point P is on the Sphere $r=a$



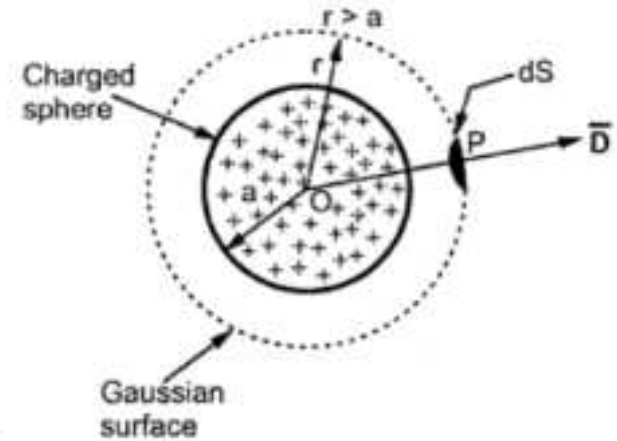
$$\vec{E} = \frac{a^3 \rho_v}{3 \epsilon_0 a^2} \vec{a}_r = \frac{\rho_v a}{3 \epsilon_0} \vec{a}_r$$

$$\vec{D} = \epsilon_0 \vec{E} = \frac{\rho_v a}{3} \vec{a}_r$$

Gauss Law to various charge distributions:

Uniformly charged sphere

Case iii) Point P inside the Sphere $r < a$



$$\vec{D} = D_r \vec{a}_r \quad \text{while} \quad d\vec{S} = r^2 \sin\theta \, d\theta \, d\phi \, \vec{a}_r$$

$$d\psi = \vec{D} \cdot d\vec{S} = D_r r^2 \sin\theta \, d\theta \, d\phi \quad \dots (\vec{a}_r \cdot \vec{a}_r = 1)$$

$$\psi = Q = \oint_S \vec{D} \cdot d\vec{S} = \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} D_r r^2 \sin\theta \, d\theta \, d\phi$$

$$= D_r r^2 [-\cos\theta]_0^{\pi} [\phi]_0^{2\pi} = 4\pi r^2 D_r$$

$$D_r = \frac{Q}{4\pi r^2}$$

$$\vec{D} = \frac{Q}{4\pi r^2} \vec{a}_r \, \text{C/m}^2$$

*Dr.M.Dhivya, Associate Professor, Department of
ECE, NHCE, Bengaluru*

Gauss Law to various charge distributions:

Uniformly charged sphere

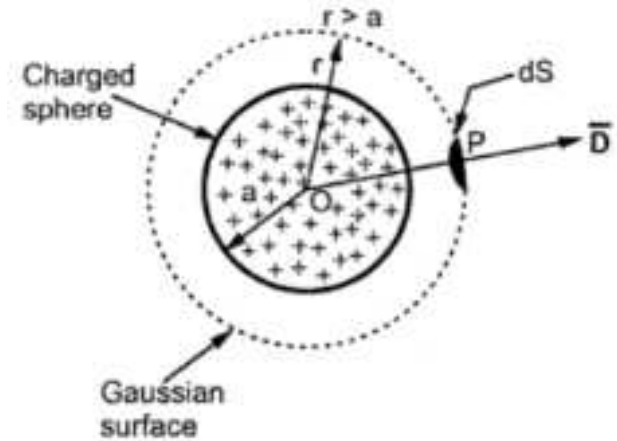
Case iii) Point P inside the Sphere $r < a$

$$Q = \int_V \rho_v dv = \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} \int_{r=0}^r r^2 \sin\theta dr d\theta d\phi$$
$$= \frac{4}{3} \pi r^3 \rho_v \quad \text{where } r < a$$

$$\bar{D} = \frac{\frac{4}{3} \pi r^3 \rho_v}{4 \pi r^2} \bar{a}_r$$

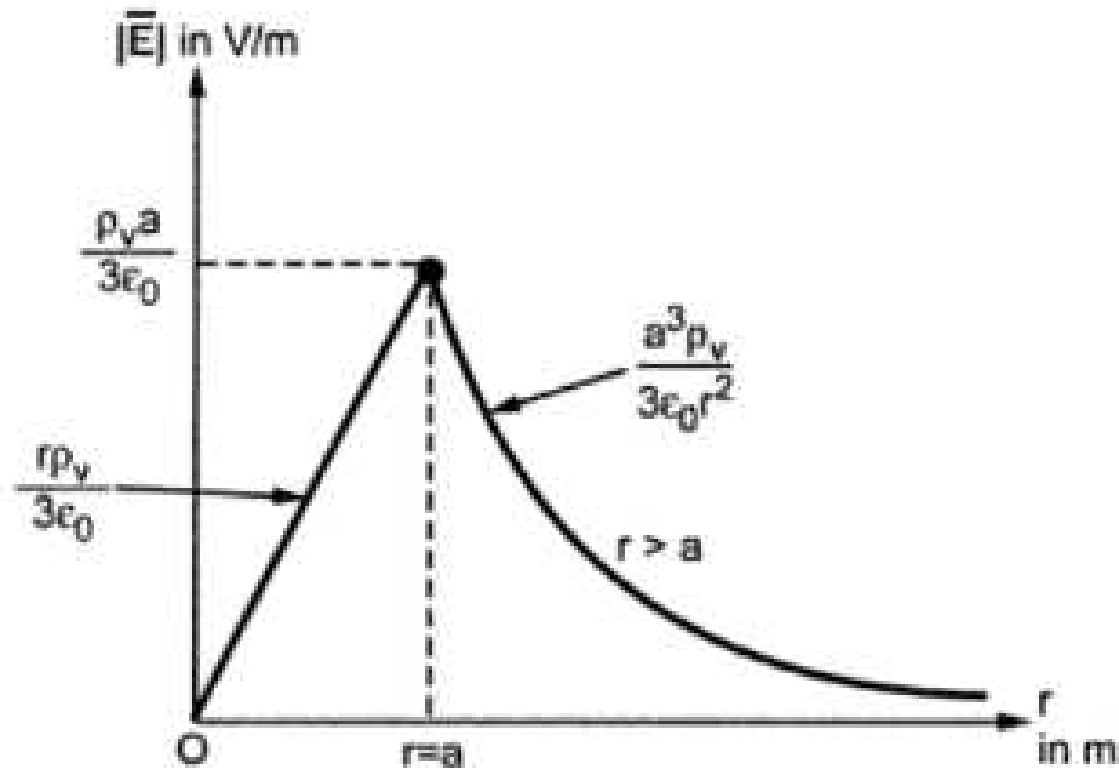
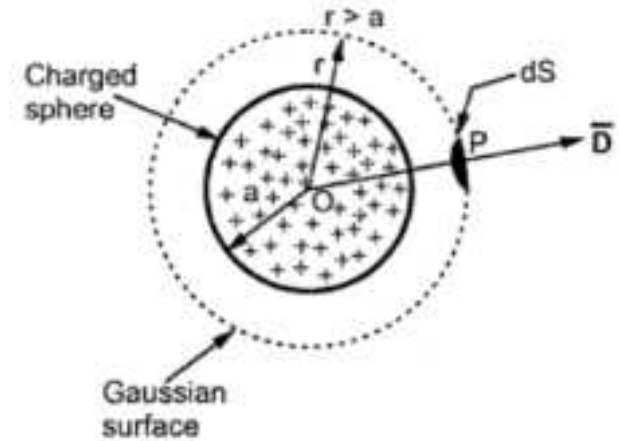
$$\bar{D} = \frac{r}{3} \rho_v \bar{a}_r \quad \text{.....} 0 < r \leq a$$

$$\bar{E} = \frac{\bar{D}}{\epsilon_0} = \frac{r}{3 \epsilon_0} \rho_v \bar{a}_r \quad 0 < r \leq a$$



Gauss Law to various charge distributions:

Uniformly charged sphere



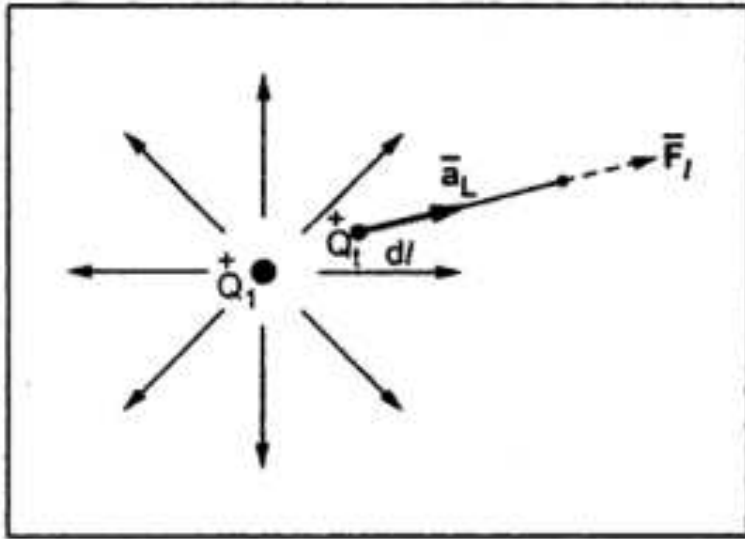
Variation of $|\vec{E}|$ against r

Energy and Potential

- **Coulombs Law** – find the electric field.
- **Gauss's law** –Determine the field about symmetrical charge arrangements.
- **Need for Electric Potential:**
 - Coulombs - Powerful than Gauss
 - Constraint – complexity of integrals for each components
 - **Solution – Scalar function with single integration.**
 - Scalar quantity, can be used to obtain electric field intensity $\vec{E}$.
 - Related parameters:
 - Potential difference,
 - relation between field intensity and electric potential,
 - potential gradient

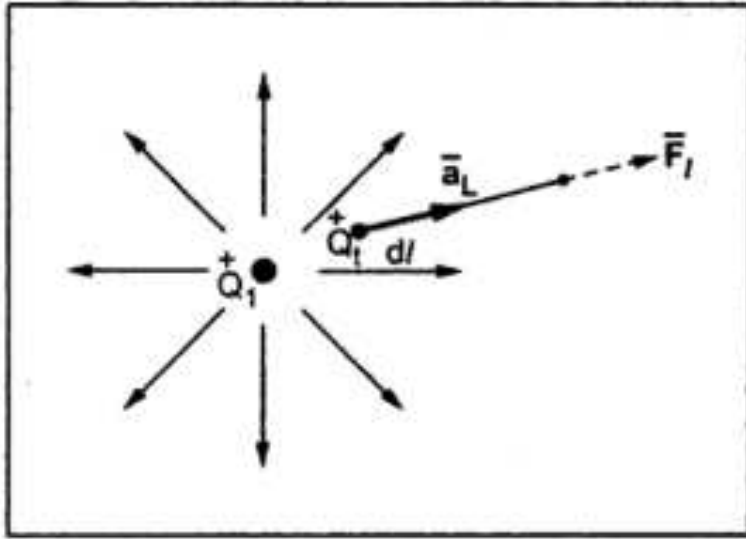
Energy expended in moving a point charge in an Electric field:

Electric field intensity: force on a unit test charge at that point, at which the value of vector field is found. $E = \frac{F}{Q}$



Repulsive force, expend the energy,
work done = potential energy of test charge Q_t , at which the point is moved.

Energy expended in moving a point charge in an Electric field:



movement of charge is dl .
 direction in which movement is
 taken place is given by $\vec{a}_L$,

Force

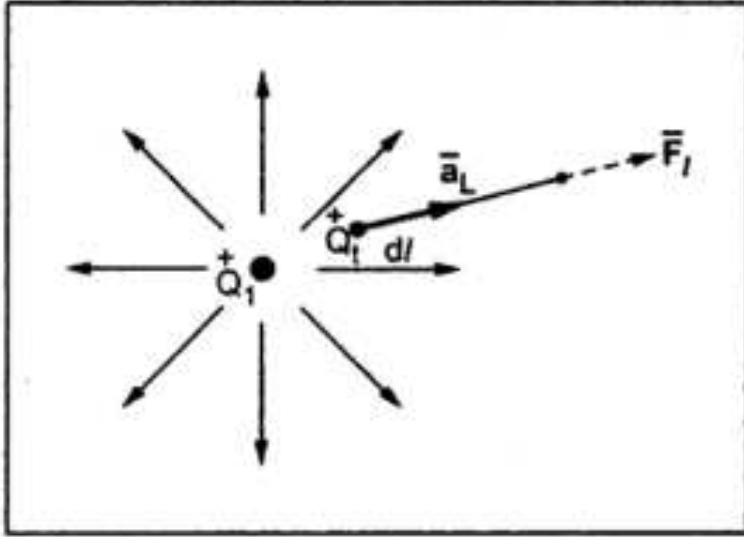
$$\vec{F} = Q_2 \vec{E} \quad \text{N}$$

Component of F in direction of $\vec{a}_L$ (dot Product)

$$\vec{F}_1 = \vec{F} \cdot \vec{a}_L = Q_2 \vec{E} \cdot \vec{a}_L \quad \text{N}$$

$$\vec{F}_{\text{applied}} = -\vec{F}_1 = -Q_2 \vec{E} \cdot \vec{a}_L \quad \text{N}$$

Energy expended in moving a point charge in an Electric field:



To Keep the charge in equilibrium, it is necessary to apply the force which is equal and opposite to the force exerted by the field in the direction of dl .

expenditure of energy is given by product of force and distance.

$$dW = \bar{F}_{\text{applied}} \times dl = -Q_t \bar{E} \cdot \bar{a}_L dl$$

$$dl \bar{a}_L = d\bar{L} = \text{distance vector}$$

$$dW = -Q_t \bar{E} \cdot d\bar{L} \text{ J}$$

dW is a scalar quantity, as $E \cdot dl$ is dot product.

Thus if a charge Q is moved from initial position to the final position, against the direction of electric field $\vec{E}$ then the total work done is obtained by integrating the differential work done over the distance from initial position to the final position.

$$W = \int_{\text{Initial}}^{\text{Final}} dW = \int_{\text{Initial}}^{\text{Final}} -Q \vec{E} \cdot d\vec{L}$$

$$W = -Q \int_{\text{Initial}}^{\text{Final}} \vec{E} \cdot d\vec{L} \text{ J}$$

Line Integral:

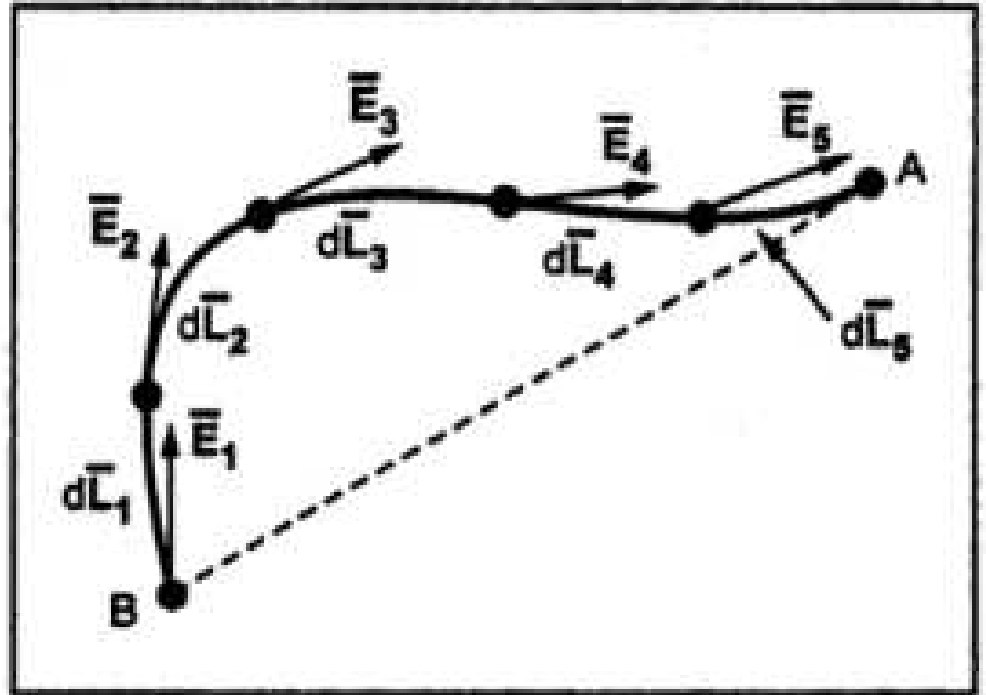
Consider that the charge is moved from initial position B to the final position A, against the electric field $\vec{E}$ then the work done is given by,

$$W = -Q \int_B^A \vec{E} \cdot d\vec{L}$$

This is called Line Integral.

Note: Line Integral is basically the summation and accurate result obtained when the number of segments become infinite.

$$W = -Q [\vec{E}_1 \cdot d\vec{L}_1 + \vec{E}_2 \cdot d\vec{L}_2 + \dots + \vec{E}_5 \cdot d\vec{L}_5]$$



Line Integral:

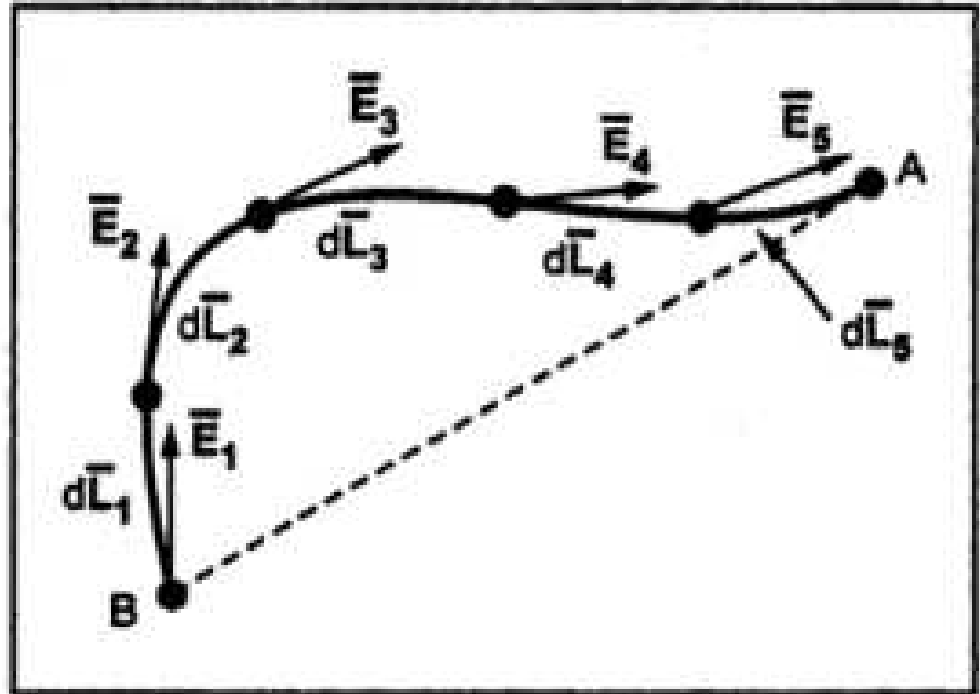
Electric field is uniform and is equal in all directions.

$$\vec{E}_1 = \vec{E}_2 = \vec{E}_3 = \vec{E}_4 = \vec{E}_5 = \vec{E}$$

$$W = -Q\vec{E} \cdot [d\vec{L}_1 + d\vec{L}_2 + \dots + d\vec{L}_5]$$

$$W = -Q\vec{E} \cdot \vec{L}_{BA} \quad \dots \text{(Uniform } \vec{E} \text{)}$$

Note: Work Done Depends on Q, E and L_{BA} (both for uniform and non-uniform electric field)



Differential Length vector $d\vec{L}$	
Cartesian $\Rightarrow$	$d\vec{L} = dx\vec{a}_x + dy\vec{a}_y + dz\vec{a}_z$
Cylindrical $\Rightarrow$	$d\vec{L} = dr\vec{a}_r + r d\phi\vec{a}_\phi + dz\vec{a}_z$
Spherical $\Rightarrow$	$d\vec{L} = dr\vec{a}_r + r d\theta\vec{a}_\theta + r \sin\theta d\phi\vec{a}_\phi$

Definition of Potential Difference and Potential:

Electric potential at a point is defined as the amount of work done in moving a unit positive charge from infinity to that point.

Potential Difference (V): work done in moving a point charge from one point to another in an electric field,

$$\text{Potential difference} = V = - \int_{\text{init}}^{\text{final}} \mathbf{E} \cdot d\mathbf{L}$$

Potential difference is measured in joules per coulomb, for which the *volt* is defined as a more common unit, abbreviated as V. Hence the potential difference between points *A* and *B* is

$$V_{AB} = - \int_B^A \mathbf{E} \cdot d\mathbf{L} \quad \text{V}$$

and V_{AB} is positive if work is done in carrying the positive charge from *B* to *A*.

The Potential field of a point charge:

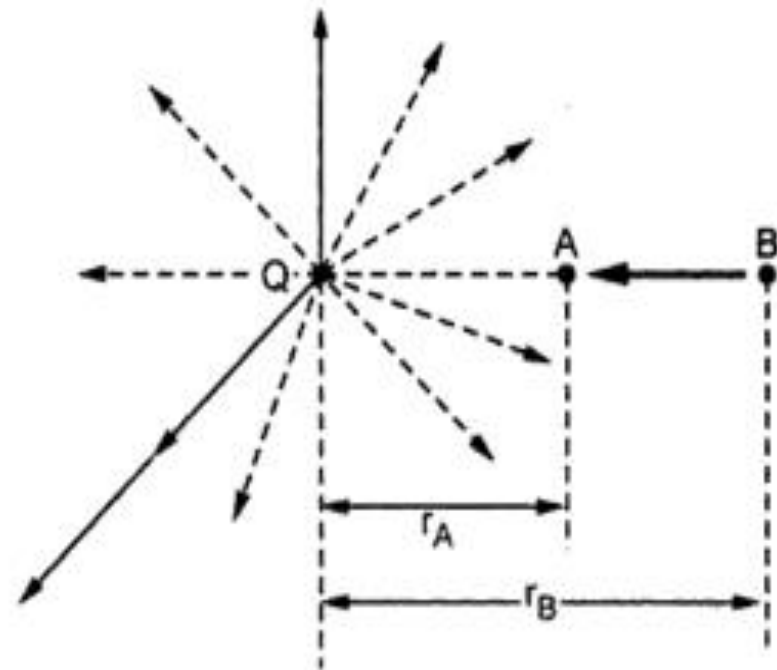
Consider a point charge, located at the origin of a spherical co-ordinate system producing $\vec{E}$ radially in all the directions

Assuming free space, the field $\vec{E}$ due to a point charge Q at a point having radial distance r from origin is given by,

$$\vec{E} = \frac{Q}{4\pi\epsilon_0 r^2} \vec{a}_r \quad \dots (1)$$

Consider a unit charge which is placed at a point B which is at a radial distance of r_B from the origin. It is moved against the direction of $\vec{E}$ from point B to point A. The point A is at a radial distance of r_A from the origin.

The differential length in spherical system is,



Potential due to a point charge Q

$$d\vec{L} = dr \vec{a}_r + r d\theta \vec{a}_\theta + r \sin \theta d\phi \vec{a}_\phi \quad \dots (2)$$

The Potential field of a point charge:

Hence the potential difference V_{AB} between points A and B is given by,

$$V_{AB} = -\int_B^A \vec{E} \cdot d\vec{L} \quad \text{But } B \Rightarrow r_B \quad \text{and} \quad A \Rightarrow r_A$$

$$\therefore V_{AB} = -\int_B^A \left(\frac{Q}{4\pi\epsilon_0 r^2} \vec{a}_r \right) \cdot (dr \vec{a}_r + r d\theta \vec{a}_\theta + r \sin\theta d\phi \vec{a}_\phi)$$

$$\therefore V_{AB} = -\int_B^A \frac{Q}{4\pi\epsilon_0 r^2} dr \quad \dots (3)$$

$$\begin{aligned} \therefore V_{AB} &= -\frac{Q}{4\pi\epsilon_0} \int_B^A r^{-2} dr = \frac{-Q}{4\pi\epsilon_0} \left[\frac{r^{-1}}{-1} \right]_B^A \\ &= -\frac{Q}{4\pi\epsilon_0} \left[-\frac{1}{r} \right]_B^A = \frac{-Q}{4\pi\epsilon_0} \left[-\frac{1}{r_A} - \left(-\frac{1}{r_B} \right) \right] \end{aligned}$$

$$\therefore \boxed{V_{AB} = -\frac{Q}{4\pi\epsilon_0} \left[-\frac{1}{r_A} + \frac{1}{r_B} \right] = \frac{Q}{4\pi\epsilon_0} \left[\frac{1}{r_A} - \frac{1}{r_B} \right] \text{ V}} \quad \dots (4)$$

when $r_B > r_A$, $\frac{1}{r_B} < \frac{1}{r_A}$ and V_{AB} is positive. This indicates the work is done by external source in moving unit charge from B to A.

Tutorial

Given the electric field $\mathbf{E} = \frac{1}{z^2}(8xyz\mathbf{a}_x + 4x^2z\mathbf{a}_y - 4x^2y\mathbf{a}_z)$ V/m, find the differential amount of work done in moving a 6-nC charge a distance of $2\text{ }\mu\text{m}$, starting at $P(2, -2, 3)$ and proceeding in the direction $\mathbf{a}_L =$: (a) $-\frac{6}{7}\mathbf{a}_x + \frac{3}{7}\mathbf{a}_y + \frac{2}{7}\mathbf{a}_z$; (b) $\frac{6}{7}\mathbf{a}_x - \frac{3}{7}\mathbf{a}_y - \frac{2}{7}\mathbf{a}_z$; (c) $\frac{3}{7} + \frac{6}{7}\mathbf{a}_y$.

Ans. $-149.3; 149.3; 0\text{ fJ}$